Data Science

Why Data Science Matters

Understanding the numbers behind the world's most valuable resource

Data is the new oil — but Data Science is the refinery

Why Data Science Matters

The Scale of Data Today :

90%

of the world's data was created in just the last 3–4 years

2.5 quintillion

bytes of data are generated every single day

Every 2–3 sec

the person generates data without realizing it

1 minute

of social uploads now exceeds entire 1990s library archives

Place Name	Numbers	Zeros
Thousands	1,000	3
Millions	1,000,000	6
Billions	1,000,000,000	9
Trillions	1,000,000,000,000	12
Quadrillion	1,000,000,000,000,000	15
Quintillion	**1,000,000,000,000,000,000**	**18**

Data isn't just growing — it's exploding at a pace humans can barely track

Why Data Science Matters

Data-Driven Business Impact

What companies are doing with all this data

Netflix
saved using recommendation algorithms that predict what you'll watch next — **$1B+ / year**

Amazon
come from data-driven product recommendations — **~35% of sales**

Self-driving cars
of sensor data processed to make real-time driving decisions — **Terabytes / hour**

Companies that use data well make more money and build better products.

Why Data Science Matters

The Challenge Hiding in the Data

More data doesn't automatically mean more value

23x

more likely to acquire customers with data-driven decisions

95%

of businesses struggle to extract value from the data they already have

~80%

of a data scientist's time is spent cleaning data, not building models

Millions

lost every year by companies due to poor data quality

Having data isn't the hard part — using it correctly is

Why Data Science Matters

Why Data Science Matters

In simple words: turning raw data into real decisions

Raw Data		Data Science		Smart Decisions
Crude oil - everywhere, but unusable as-is	→	The refinery - cleans, structures, and finds patterns	→	Fuel for growth - better products, less waste

- We're drowning in data → every click, purchase, and sensor reading adds to the pile, all day, every day.
- Most of it is messy and unusable as-is → that's why so much effort goes into cleaning it before anything useful can be done.
- Data science finds the patterns humans can't see → like what you'll want to buy next, or when a machine will break down.
- Companies that do this well win → better recommendations, fewer wasted resources, more value created.

Why Data Science Matters

Data Science in Industry — Trends

What's actually changing on the ground, not just in headlines

- **80% of enterprise AI failures** are due to poor data quality, not algorithms
- **AutoML tools** are reducing model-building time, increasing focus on data preperation
- **Model monitoring & refresh** is now mandatory in BFSI and healthcare
- **Explainable AI (XAI)** is becoming essential due to regulations
- **Demand is shifting** from “model builders” to “end-to-end data problem solvers”

Why Data Science Matters

Job Market Insight

What separates a hired candidate from an overlooked resume

Employers expect:

Problem framing
translating a business question into a data question

Data cleaning
the unglamorous work that actually decides model quality

Business understanding
knowing why the analysis matters to the company

Algorithms = **baseline skill**

not a differentiator

Everyone can call scikit-learn. Few can frame the right problem and clean the data well enough to trust the answer.

"Understanding the problem matters more than the algorithm."

Why Data Science Matters

Real-World Impact — Three Industries

Data science shows up very differently depending on the problem

RETAIL

The Gap

Data science drives pricing, inventory levels, and shipping decisions. Cut price-elasticity calculations from days to hours.

spectrum.ieee.org/gap-data-science-director

SOCIAL GOOD

NYU VIDA

Traces one known wildlife-trafficking website outward with ML to map hidden trafficking networks for law enforcement.

spectrum.ieee.org/nyu-vida-data-science

HEALTHCARE

Pfizer

A pre-built data pipeline let COVID vaccine trial data move through review 4x faster speed came from data infrastructure, not a smarter model.

spectrum.ieee.org/pfizer-covid-vaccines-data-science

Why Data Science Matters

WHAT DATA SCIENCE DOES

A retail case study: how The Gap turns raw transactions into pricing, inventory, and shipping decisions

- **Predicting Demand**

  Analyzes consumer habits to forecast what products people will want.

- **Setting Prices**

  Price-elasticity models balance sales volume against margin.

- **Planning Manufacturing**

  Data guides how many units of each item should be produced.

- **Optimizing Logistics**

  Finds more efficient routes and methods for moving goods.

Hours
not days, for pricing calcs

Demand
forecast from real behavior

Inventory
optimized by algorithm

RETAIL CASE STUDY

The Gap

Vivek Anand, Director of Data Science, leads a team of statisticians and operations researchers turning transaction data into pricing, inventory, and shipping decisions.

“Data science solves problems that were previously unsolvable - grouping transactions that look different on the surface but are similar underneath.”

Result: price-elasticity calculations cut from days to hours

Why Data Science Matters

Fighting Wildlife Trafficking With Data Science

A research case study: how NYU Tandon hunts the digital fingerprints of a multi-billion-dollar illegal trade

HOW DATA SCIENCE HELPS

- **Mapping Trafficker Networks**
  Tracks digital fingerprints across sites to reveal hidden trafficking networks.
- **Predictive Modeling**
  Machine learning models keep improving, surfacing new trafficking hubs over time.
- **Deep-Web Search**
  Builds a specialized search engine that reaches where standard search can't.
- **Powering Law Enforcement**
  Delivers actionable leads to agencies with too little bandwidth to track traffickers alone.

RESEARCH CASE STUDY

NYU Tandon — VIDA

Juliana Freire, Professor of Computer Science & Data Science, directs VIDA and leads a four-expert team spanning data science, environmental studies, and criminal justice.

"Animal trafficking has many dangers, not least to the animals. We can use data science to help fight this criminal enterprise."

Result: $994,000 NSF (National Science Foundation) grant now funds the effort

Why Data Science Matters

How Data Science Powered Pfizer's COVID-19 Vaccine

A pharma case study: how an informatics team turned clinical trial samples into an FDA green light in under a year

HOW DATA SCIENCE HELPED

Tracking Clinical Samples

A LIMS platform received, tracked, and divided every blood sample from trial participants.

Automating Complex Analysis

Robotics and instruments ran assays like PCR and virus-neutralization tests at scale.

Statistical Modeling

SAS and R algorithms turned raw assay results into statistically validated data.

Powering Regulatory Approval

Aggregated, validated data was reported directly to the FDA for review.

PHARMA CASE STUDY

Pfizer — COVID-19 Vaccine

Frank DePierro leads Pfizer's informatics and digital technology team, turning clinical trial data into an FDA-cleared vaccine in under a year.

"The tools that track all of those samples and generate the data — that's my team."

Result: more COVID samples logged in 18 months than all prior vaccine programs combined since 2014

Module-01

Introduction to Data Science

Syllabus Contents

1.1 Introduction to Data Science

1.2 Motivation to use Data Science Techniques: Volume, Dimensions and Complexity

1.3 Data Science Tasks and Examples

1.4 Data Science Process, Model building, Cross Validation, Bootstrapping

1.5 Difference between data science and data analytics

1.6 Applications of Data Science

Syllabus Contents

1.1 Introduction to Data Science

LO 1.2-Apply engineering fundamentals to explain the core concepts of Data Science, its overall process, and the motivation for adopting data-driven approaches. (PI 1.3.1)

Reference: Vijay Kotu, Bala Deshpande. Data Science Concepts and Practice, Elsevier, M.K. Publishers.

Introduction to Data Science

What Is Data Science?

The Core Idea

"science" signals that the methods are evidence-based - built on empirical knowledge, and more specifically, historical observations.

Purpose: turn raw data into actionable, decision-ready insight.

Data Science is an interdisciplinary field involving collecting, processing, storing, analyzing, and interpreting large volumes of structured and unstructured data to extract meaningful insights and knowledge.

ALSO KNOWN AS

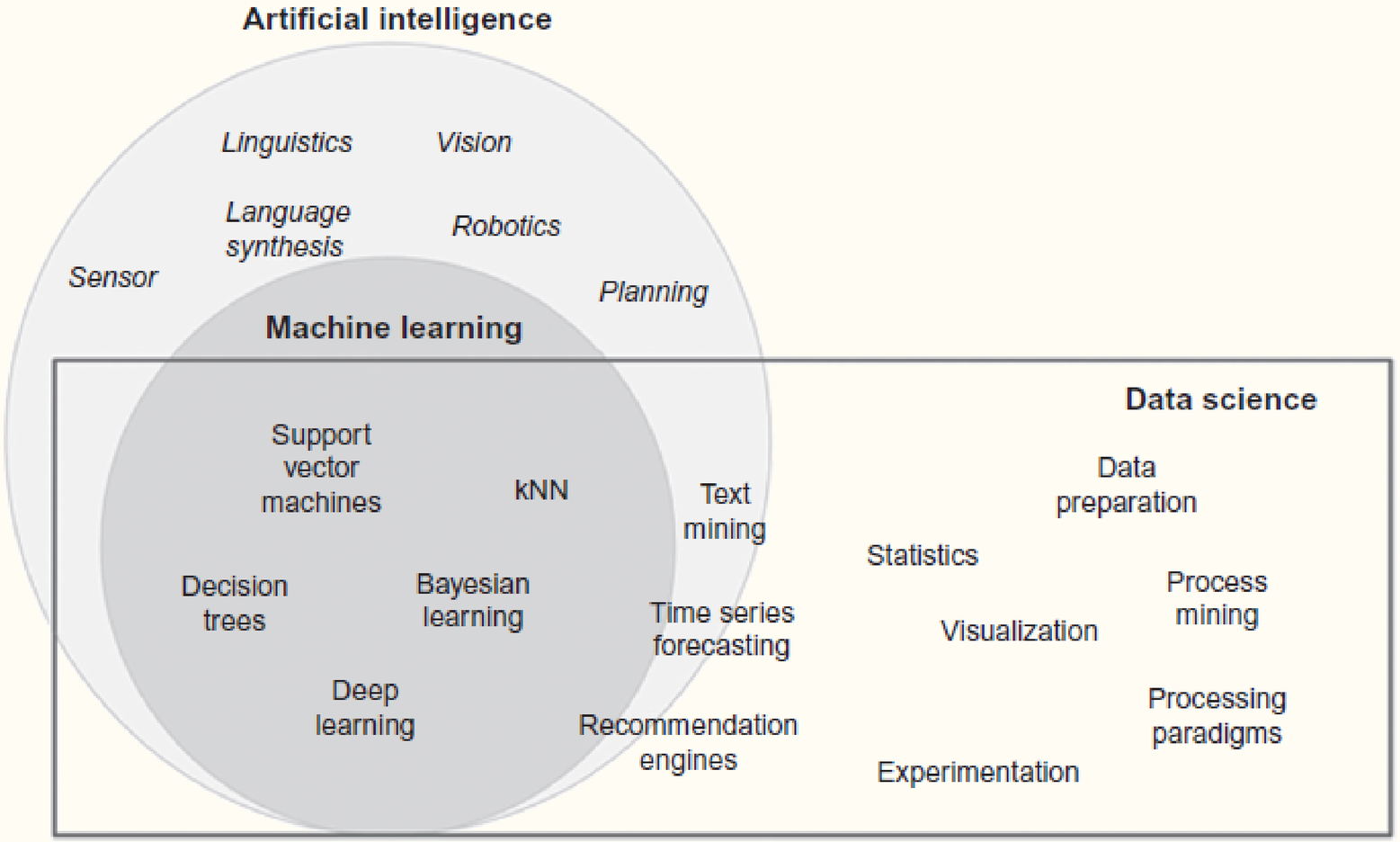
- Knowledge Discovery
- Machine Learning
- Predictive Analytics
- Data Mining

Each term carries a slightly different connotation depending on context.

Introduction to Data Science

AI, Machine Learning & Data Science

- *Related, often conflated, but distinct - three nested fields with their own scope.*



Artificial Intelligence

Giving machines the capability of mimicking human cognitive behavior.

Machine Learning

Giving machines the capability of learning from data (experience).

Data Science

The business application of AI, ML, statistics & visualization.

Introduction to Data Science

Artificial Intelligence

- *Giving machines the capability of mimicking human behavior particularly cognitive functions.*

Where It Shows Up

- Facial recognition
- Automated driving
- Sorting mail by postal code

In some tasks machines have far exceeded human capability (sorting thousands of pieces of mail in seconds); in others, we've barely scratched the surface.

TECHNIQUES UNDER AI

Linguistics	**Natural Language Processing**	**Decision Science**
Vision	**Robotics**	**Planning**

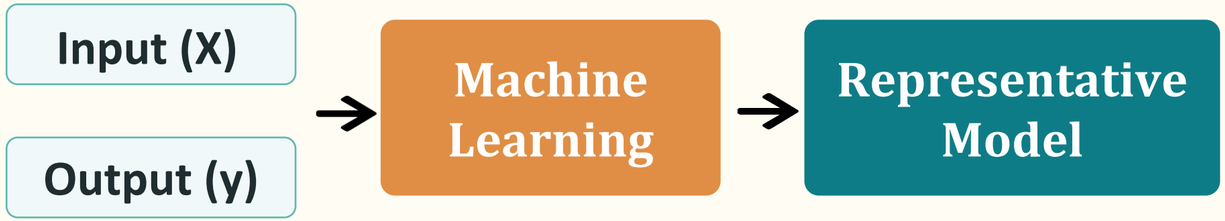
Introduction to Data Science

Machine Learning

- *A sub-field or tool of AI: giving machines the capability of learning from experience.*



Traditional: rules transform input into output.



ML: known input + output (training data) teach the model.

Worked Example: Content Moderation

Shown labeled examples of abusive vs. non-abusive posts, a “learner” generalizes a pattern

ALGORITHMS UNDER ML

- Support Vector Machines
- k-NN
- Decision Trees
- Bayesian Learning
- Deep Learning

Introduction to Data Science

What is Data Science?

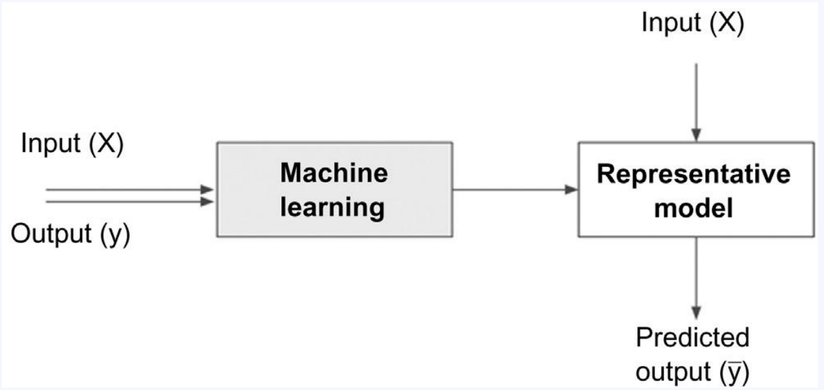
Data science starts with data from a simple array of numbers to millions of observations with thousands of variables and uses specialized methods to discover meaningful structure.

Extracting Meaningful Patterns

- Identifies valid, novel, and useful patterns to support decisions.
- Involves inference and iteration over many hypotheses.
- Patterns must generalize to new, unseen data not just the original dataset.

Building Representative Models

- A model represents the relationship between variables in a dataset.
- Serves two purposes: prediction for new inputs, and explanation of relationships.
- e.g. credit score, income, and loan amount used to model interest rate.



Introduction to Data Science

Foundations & Learning Algorithms

Data science combines statistical and computational disciplines with automated, iterative learning methods.

Statistics, ML & Computing

- Combines techniques from statistics, machine learning, and database theory.
- Subject matter expertise is essential to interpret data and business context.
- An iterative process, often run on large, distributed datasets.

Learning Algorithms

- Tasks: classification, regression, clustering, association analysis.
- Algorithms: decision trees, neural networks, k-NN, k-means.

Introduction to Data Science

Associated Fields

Data science relies on several established techniques throughout its process.

Descriptive Statistics

Mean, standard deviation, and correlation summarize dataset structure.

Hypothesis Testing

Confirmatory analysis to validate hypotheses, e.g. A/B testing.

Exploratory Visualization

Visualizing data to reveal patterns and relationships.

Data Engineering

Sourcing, storing, and preparing data for analysis.

Dimensional Slicing (OLAP)

Slicing and filtering data across dimensions and measures.

Business Intelligence

Distributing insights and combining historical with predicted trends.

Syllabus Contents

1.2 | Motivation to use Data Science Techniques: Volume, Dimensions and Complexity

LO 1.1 - Apply the knowledge of discrete structures, linear algebra, statistics, and numerical techniques to interpret fundamental data characteristics such as volume, dimensionality, and complexity. (PI 1.1.1)

Reference: Vijay Kotu, Bala Deshpande. Data Science Concepts and Practice||, Elsevier, M.K. Publishers.

Motivation to use Data Science Techniques

WHY DATA SCIENCE, WHY NOW

01 Volume

How much data there is

The sheer amount of data captured keeps growing.

02 Dimensionality

How many attributes describe each record

Every record carries more attributes

03 Complexity

How hard the question is to answer

Extracting meaningful patterns from such data is challenging.

Motivation to use Data Science Techniques

VOLUME

The Sheer Amount of Data Is Exploding

↓ Cost

Storage & processing costs keep falling

↑ Capture

Every transaction and event is now recordable

↑ Granularity

Data becomes more detailed, record by record

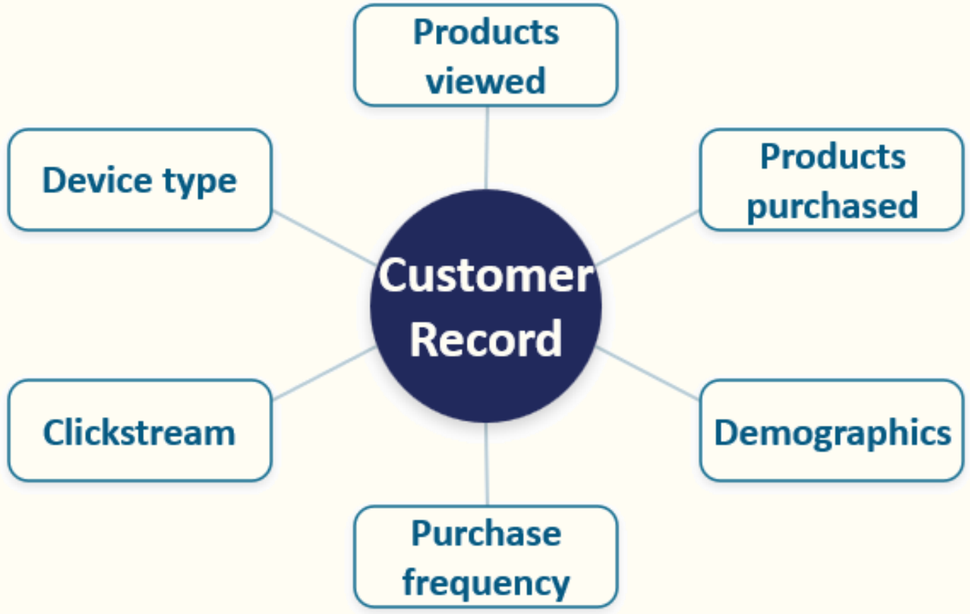
The rapid increase in data volume exposes the limits of current analysis methods. In some applications, the time needed to build a generalized model is critical — and data volume is a major factor in determining that time frame, from development through deployment.

Motivation to use Data Science Techniques

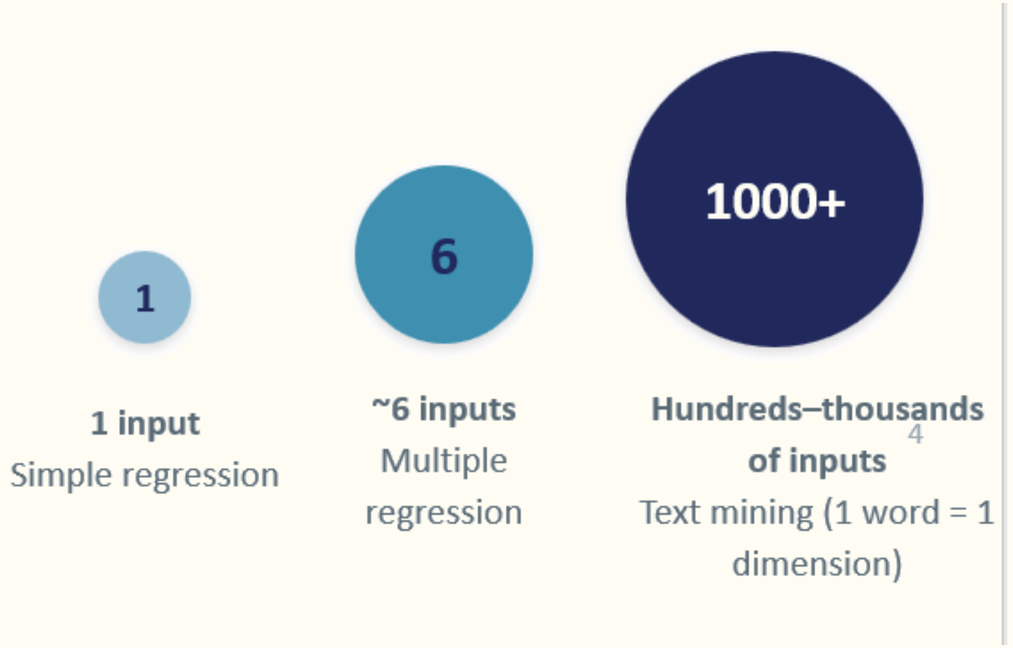
DIMENSIONS

More Attributes, More Dimensions, More Complexity

Example: one e-commerce customer record



As dimensions increase, model complexity increases



Motivation to use Data Science Techniques

COMPLEXITY

Two Ways to Ask a Question of Data

As data gets more complex, the questions we need answered get more complex too.

CONFIRMATORY

Traditional statistical analysis

Highly successful for modeling simple relationships between a response and its inputs — it validates a hypothesis you already have.

Limited when data gets large and complex.

EXPLORATORY

Machine learning

Searches for an algorithmic model that predicts the output from the inputs. It runs recursively, learning from the predictive errors of each previous cycle — generating a multitude of hypotheses rather than testing just one.

Built for large, high-dimension search spaces.

Motivation to use Data Science Techniques

“We Need Both Exploratory and Confirmatory”

John Tukey, 1980 — quoted for the idea that neither approach replaces the other.

- Volume: falling storage costs mean more data than ever needs a scalable approach
- Dimensions: more attributes per record means traditional slicing and testing don't scale
- Complex questions: today's problems need both hypothesis-testing and algorithmic exploration

Syllabus Contents

1.3 Data Science Tasks and Examples

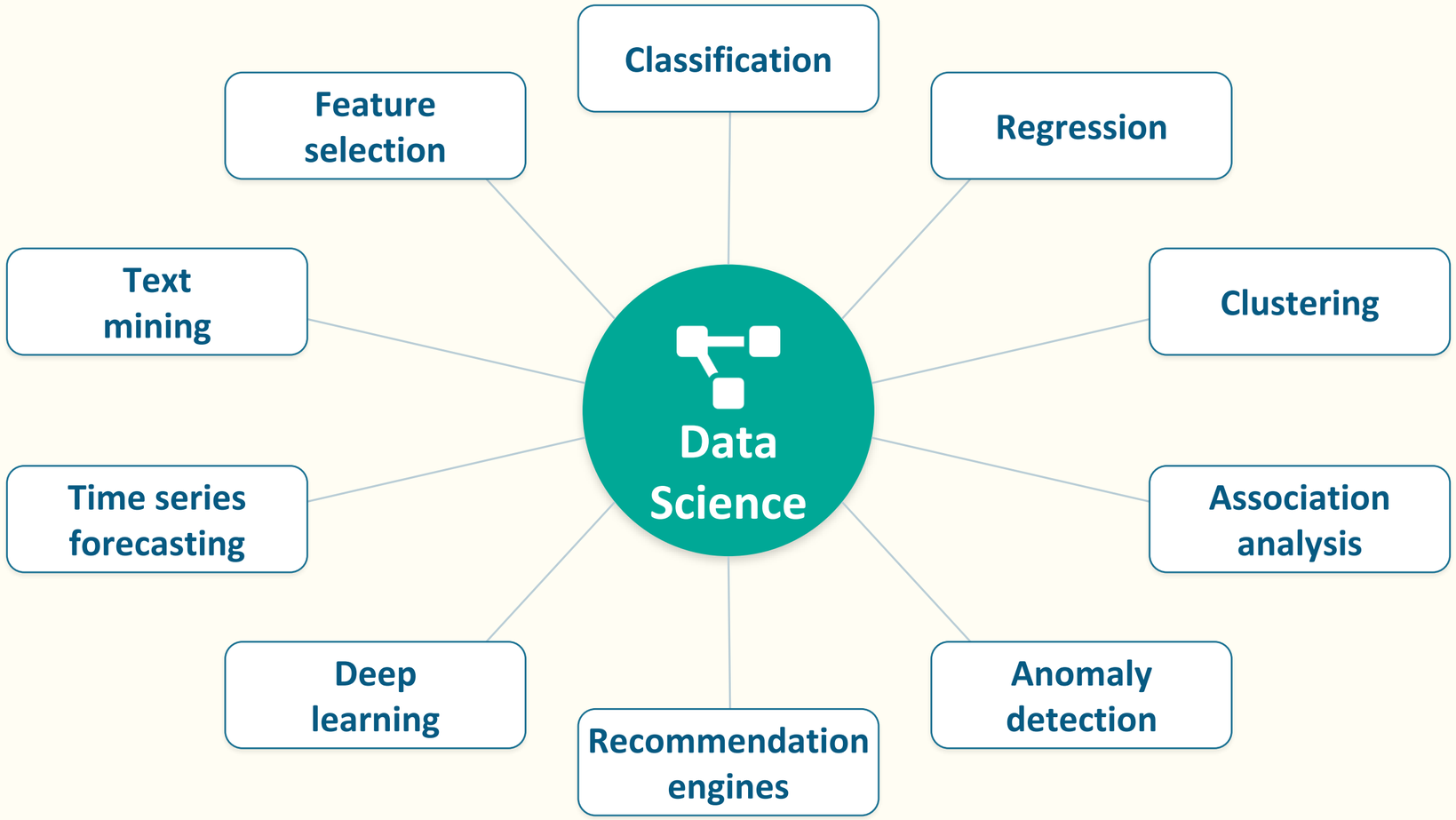
LO 1.3- Evaluate a given data-related problem statements and determine the required analysis, while continually learning and adapting to new data science tools and techniques

Reference: Vijay Kotu, Bala Deshpande. Data Science Concepts and Practice, Elsevier, M.K. Publishers.

Data Science Tasks

Data Science Problems Map to Specific Tasks

Beyond the supervised/unsupervised split, problems are further classified by the task they solve.



Data Science Tasks

What Each Task Actually Does

Tasks	Description	Algorithms	Examples
Classification	Predict if a data point belongs to one of the predefined classes. The prediction will be based on learning from a known dataset	Decision trees, neural networks, Bayesian models, induction rules, *k*-nearest neighbors	Assigning voters into known buckets by political parties, e.g., soccer moms Bucketing new customers into one of the known customer groups
Regression	Predict the numeric target label of a data point. The prediction will be based on learning from a known dataset	Linear regression, logistic regression	Predicting the unemployment rate for the next year Estimating insurance premium
Anomaly detection	Predict if a data point is an outlier compared to other data points in the dataset	Distance-based, density-based, LOF	Detecting fraudulent credit card transactions and network intrusion
Time series forecasting	Predict the value of the target variable for a future timeframe based on historical values	Exponential smoothing, ARIMA, regression	Sales forecasting, production forecasting, virtually any growth phenomenon that needs to be extrapolated
Clustering	Identify natural clusters within the dataset based on inherit properties within the dataset	*k*-Means, density-based clustering (e.g., DBSCAN)	Finding customer segments in a company based on transaction, web, and customer call data
Association analysis	Identify relationships within an item set based on transaction data	FP-growth algorithm, Apriori algorithm	Finding cross-selling opportunities for a retailer based on transaction purchase history
Recommendation engines	Predict the preference of an item for a user	Collaborative filtering, content-based filtering, hybrid recommenders	Finding the top recommended movies for a user

LOF, local outlier factor; ARIMA, autoregressive integrated moving average; DBSCAN, density-based spatial clustering of applications with noise; FP, frequent pattern.

Syllabus Contents

1.4 Data Science Process

LO 1.2-Apply engineering fundamentals to explain the core concepts of Data Science, its overall process, and the motivation for adopting data-driven approaches. (PI 1.3.1)

LO 1.3- Evaluate a given data-related problem statements and determine the required analysis, while continually learning and adapting to new data science tools and techniques

LO 1.4- Apply computer engineering principles to outline appropriate data science procedures and basic model evaluation methods such as cross-validation, K-fold, and bootstrapping, while remaining aware of ongoing technological advancements in the field. (PI 2.3.1, PI 11.2.2)

Reference: Vijay Kotu, Bala Deshpande. Data Science Concepts and Practice‖, Elsevier, M.K. Publishers.

Data Science Process

THE DATA SCIENCE FRAMEWORK

- *A methodical, iterative framework for discovering useful relationships and patterns in data*

Common Process Frameworks

CRISP-DM

Cross **I**ndustry **S**tandard **P**rocess for **D**ata **M**ining

Data-mining consortium, 2000 — most widely adopted

SEMMA

Sample, **E**xplore, **M**odify, **M**odel, **A**ssess

Developed by the SAS Institute

DMAIC

Define, **M**easure, **A**nalyze, **I**mprove, **C**ontrol

Used in Six Sigma practice

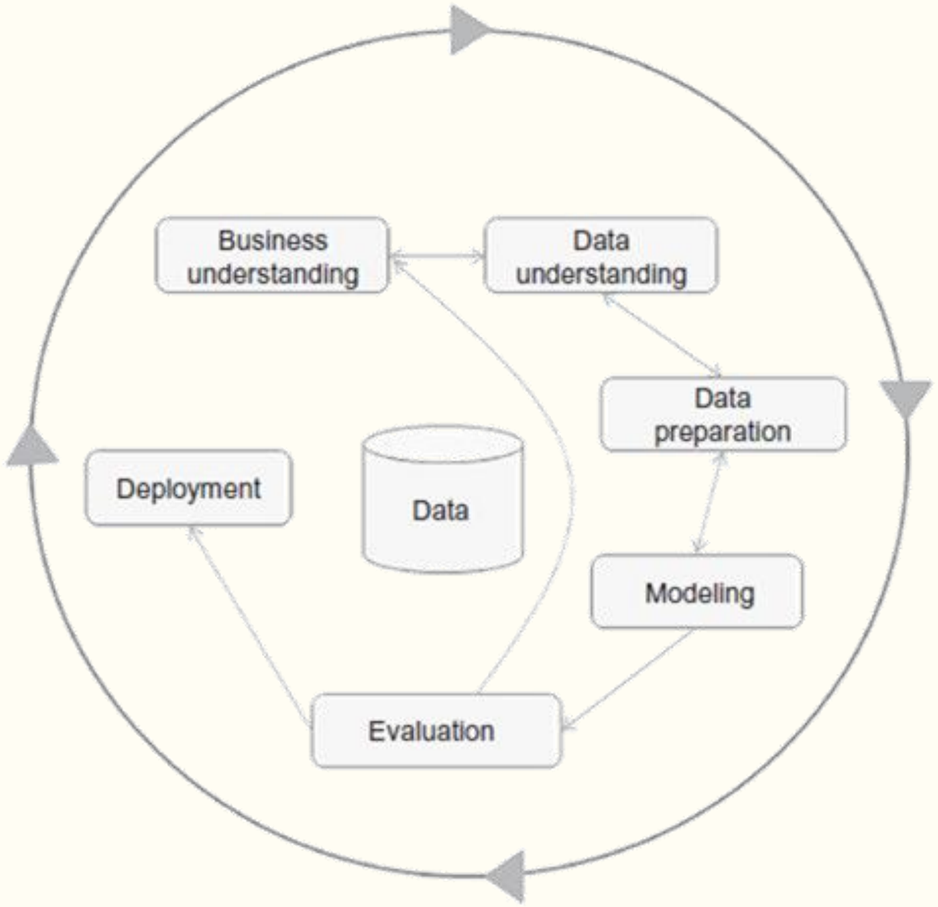
KDD Process

Selection, Preprocessing, Transformation, Data Mining, Interpretation, Evaluation

Knowledge Discovery in Databases

Data Science Process

The CRISP-DM Framework: *The most widely adopted process framework for developing data science solutions (Chapman et al., 2000).*



1

Problem Understanding

Define business goals and success criteria

2

Prepare the Data

Collect, clean, and sample the data

3

Develop the Model

Select and train candidate models

4

Apply the Model

Test performance on real-world data

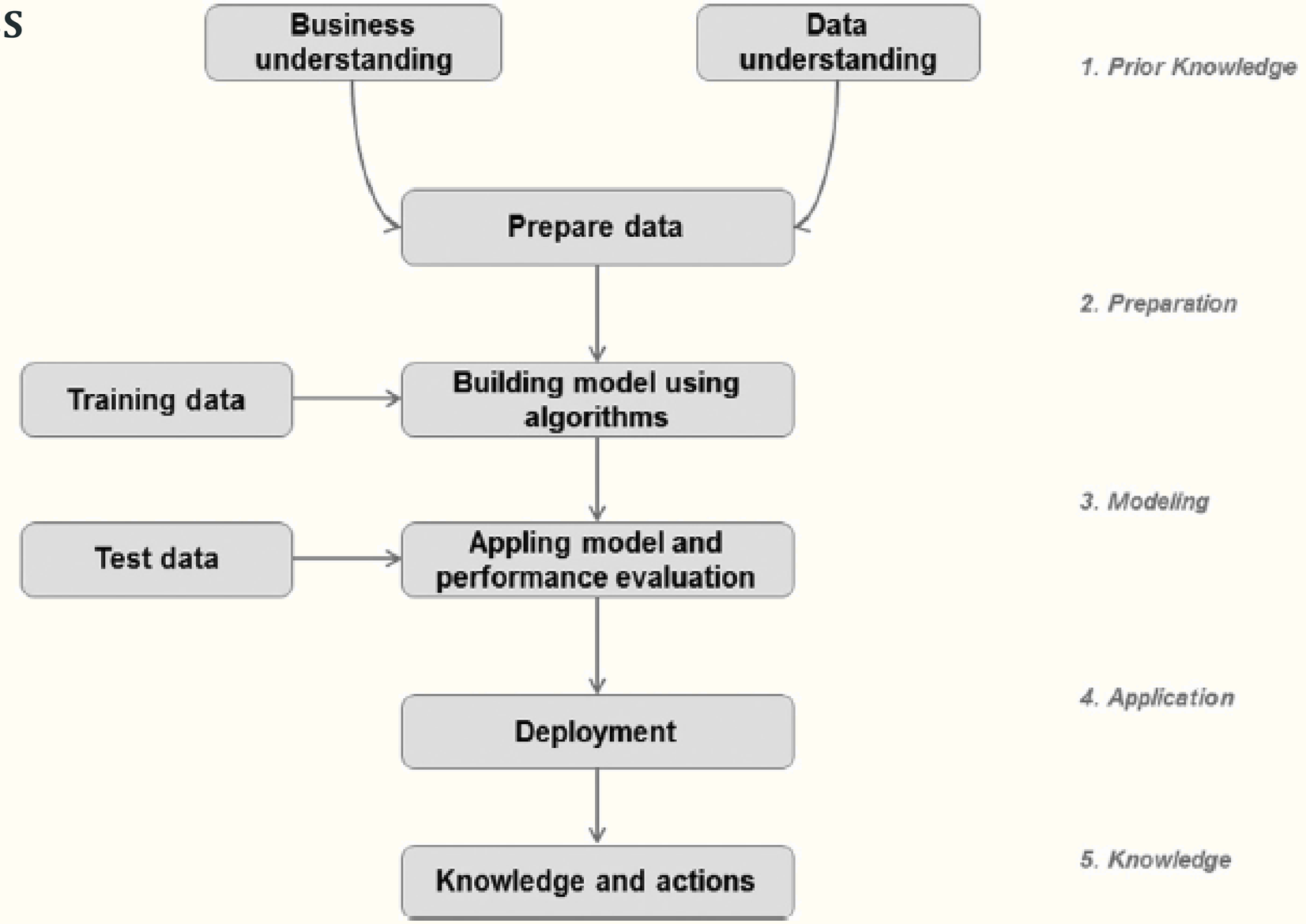
5

Deploy & Maintain

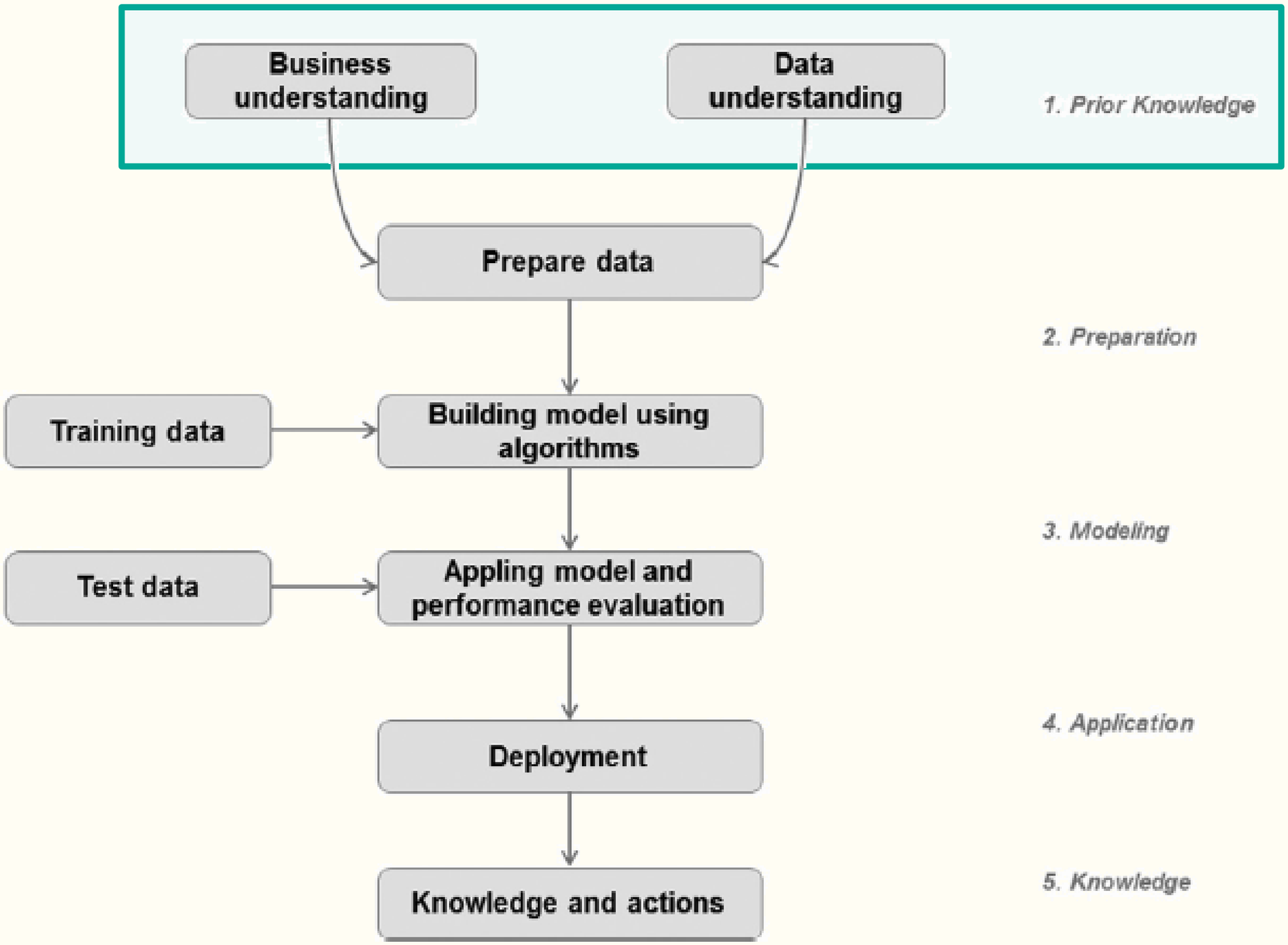
Operationalize and monitor over time

Data Science Process

THE DATA SCIENCE PROCESS



Data Science Process



Data Science Process

1. Prior Knowledge

Prior knowledge is information already known about a subject; the process begins here, before any data is touched.

What Is Prior Knowledge?

- Refers to information already known about a subject or domain.
- A data science problem does not arise in isolation — it is built on:
  - Existing subject-matter expertise
  - Business context
  - Historical understanding
- This step helps define the problem, fit it to the business context, and identify the data required.

1.1 Objective

- The process starts with a business objective or analytical question — the most important step in the lifecycle.
- **Example (Loan Prediction): banks assess loan prediction using credit score, income, and liabilities.**
- ***Objective: Can we predict the interest rate for a new borrower using past loan data?***
- A clear objective helps: identify the right data, select appropriate algorithms, and align analysis with business goals.
- Data science is iterative, but the objective must be defined correctly at the start.

"A well-defined objective leads to meaningful data science solutions."

Data Science Process

1.2 Subject Area (Domain Knowledge)

- Data science reveals many patterns, but not all are valid or meaningful — false/spurious patterns are common.
- The data scientist must identify relevant patterns and reject misleading correlations.
- This requires strong subject-area knowledge and understanding of the business process generating the data.

"Data reveals patterns, but domain knowledge decides which patterns matter."

Loan Prediction example: To predict interest rate, understanding the business means being able to answer:

- How does the business actually work?
- Why does this prediction matter — what decision does it drive?
- What happens after the rate is predicted?
- What data can be collected from borrowers?
- What data can't be — due to regulation or internal policy?
- What external factors also move the interest rate?
- How is the outcome's validity verified?

Data Science Process

1.3 Data

- Prior knowledge of data is as important as domain knowledge.
- Essential to understand how data is collected, stored, transformed, and reported/used.
- This step reviews available data and identifies missing data or gaps.
- Key factors: data quality and quantity, availability and gaps, and whether limitations require changing the business question.

A data science model is only as good as the data used to build it

Borrower ID	Credit Score	Annual Income	State	Interest Rate (%)
01	500	$42,000	NY	7.31
02	600	$58,000	CA	6.70
03	700	$71,000	TX	5.95
04	700	$71,500	TX	6.40
05	700	$71,500	TX	6.40
06	800	$95,000	FL	5.40
07	800	$102,000	FL	5.70
08	750	$88,000	New York	5.90
09	550	$39,000	CA	7.00
10	650	$61,000	TX	6.50
11	825	$110,000	FL	5.70
12	700	—	NY	6.10
13	900	$67,000	CA	5.20
14	620	$1,250,000	TX	6.60

Data Science Process

1.4 Correlation vs Causation

Correlation indicates a statistical relationship between variables; **causation** implies one variable directly influences another. Correlation does not guarantee causation.

Example: the interest rate for a new borrower is predicted using credit-score patterns from historical data.

Incorrect question

“Can a borrower’s credit score be predicted using interest rate?”

Technically possible, but business-wise incorrect.

More Examples.

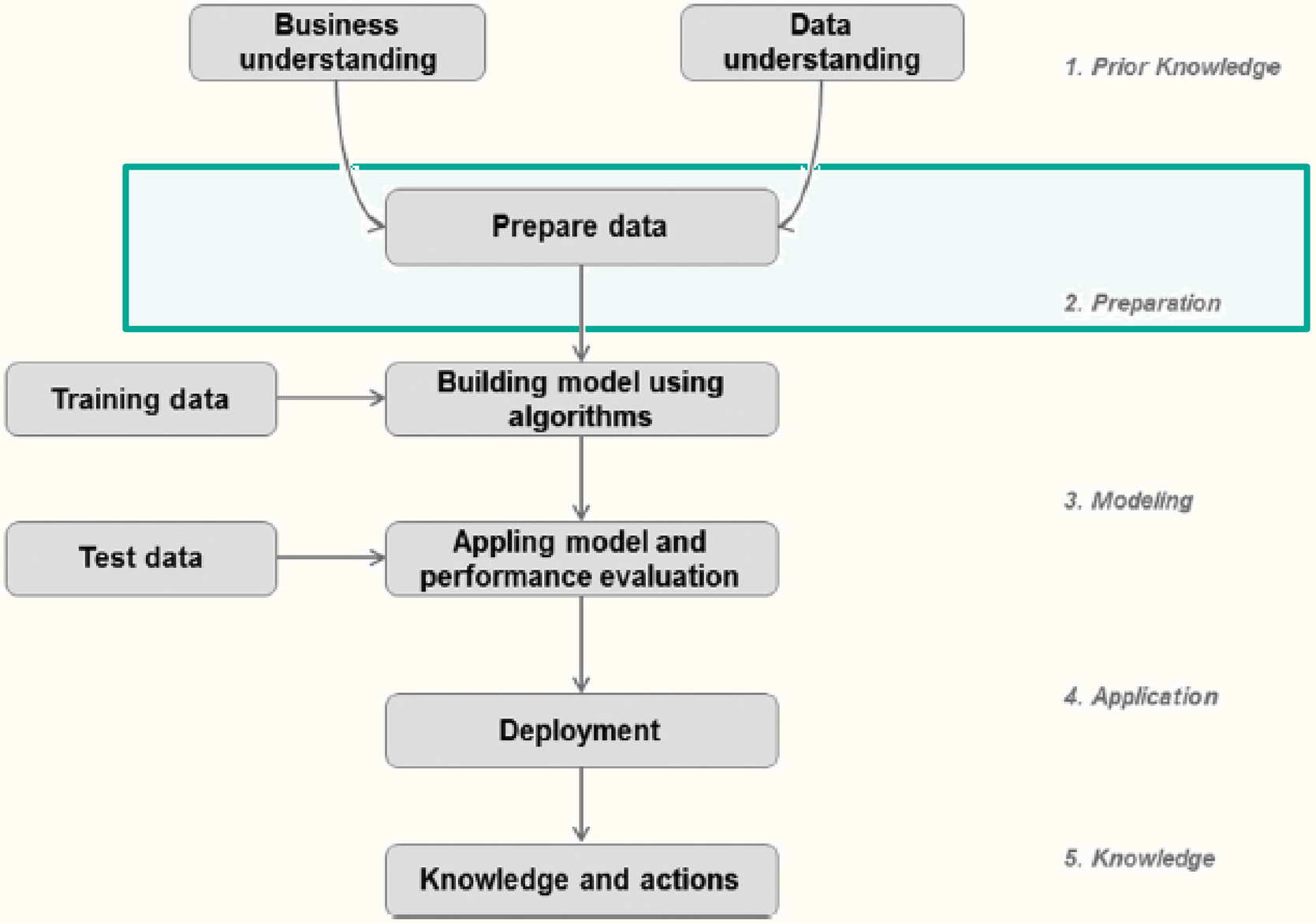
Correct causal direction

Credit score → Interest rate

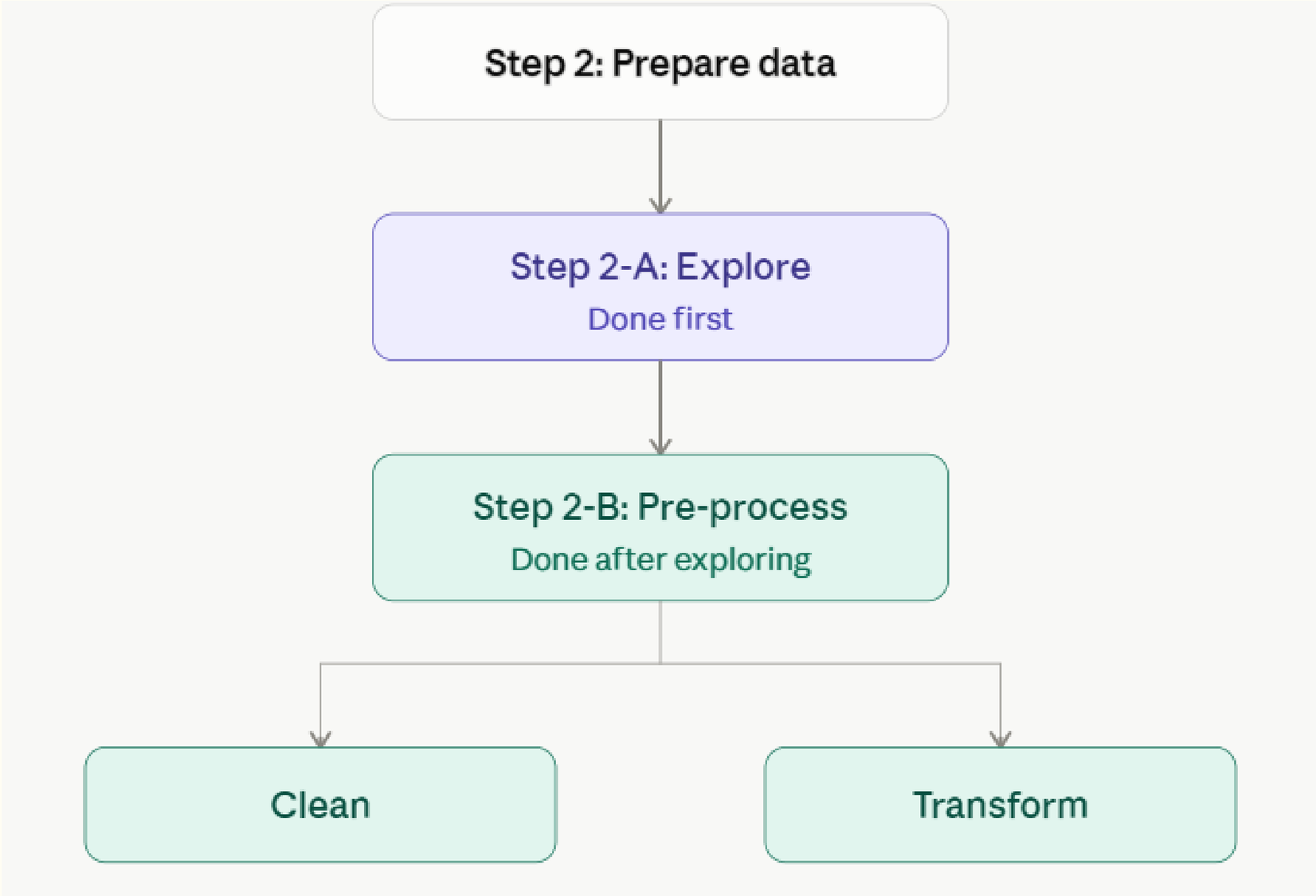
Domain knowledge helps identify correct cause-effect relationships and frame meaningful data science questions.

“Correlation can mislead; causation must guide model design.”

Data Science Process



Data Science Process



Data Science Process

Step 2.1 — Data Exploration

Descriptive statistics and a visualization expose the dataset's structure — and hint that something is off.

Descriptive Statistics (raw, 14 rows)

Field	Mean	Median	Std Dev
Credit Score	699.6	700	110.8
Interest Rate (%)	6.20	6.25	0.60
Annual Income	$163,539	$71,500	large

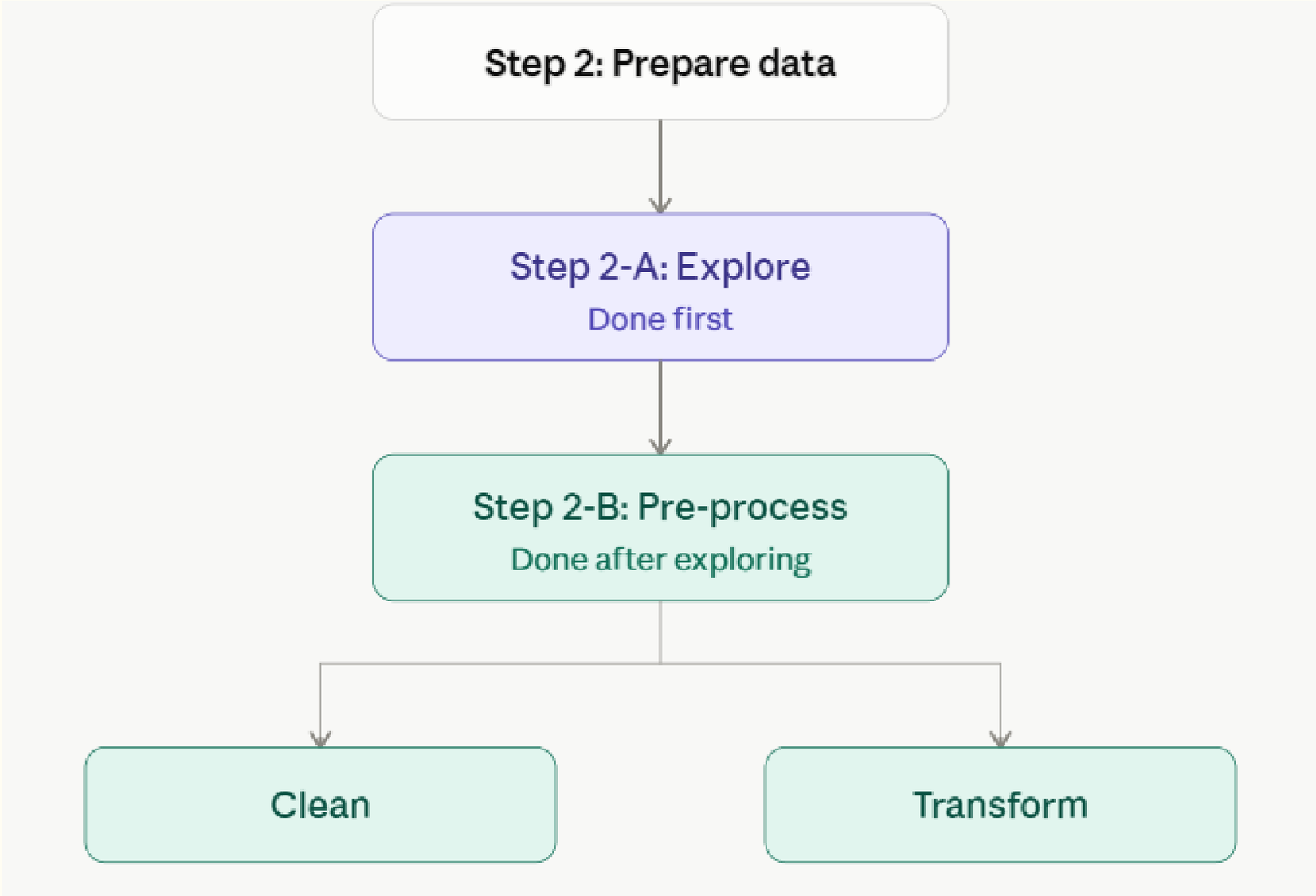
- *Mean and median are close together so no skew there individually*
- *The income mean ($163,539) is more than double its median ($71,500) — a red flag that a single extreme value is dragging the average up. That's Borrower 13.*

Scatterplot: Credit Score vs. Interest Rate



As credit score rises, interest rate falls — the trend we expect. One credit score (900) is off the valid 300–850 scale (standard)

Data Science Process



Data Science Process

Step 2.2 — Data Preparation (cleaning)

Real-world data is messy — exploration is how these problems get found before modeling.

Inconsistent values

Same fact recorded different ways (e.g. “NY” vs. “New York”).

Duplicate records

The same entity appears more than once, inflating counts and skewing stats.

Missing values

Gaps in the data — identified during exploration, fixed during preparation.

Invalid data

Impossible values, like a credit score of 900 or a negative age.

Outliers

Extreme values that may be errors, or the very thing you're modeling for.

Data Science Process

Step 2.2 — Data Preparation (cleaning)

Real-world data is messy — exploration is how these problems get found before modeling.

Borrower ID	Credit Score	Annual Income	State	Interest Rate (%)
01	500	$42,000	NY	7.31
02	600	$58,000	CA	6.70
03	700	$71,000	TX	5.95
04	**700**	**$71,500**	**TX**	**6.40**
04	**700**	**$71,500**	**TX**	**6.40**
05	800	$95,000	FL	5.40
06	800	$102,000	FL	5.70
07	**750**	**$88,000**	**New York**	**5.90**
08	550	$39,000	CA	7.00
09	650	$61,000	TX	6.50
10	825	$110,000	FL	5.70
11	**700**	**—**	**NY**	**6.10**
12	**900**	**$67,000**	**CA**	**5.20**
13	**620**	**$1,250,000**	**TX**	**6.60**

■ **Highlighted rows contain a data-quality problem — spotted next in Data Exploration and named in Step 2.**

5 issues are hiding in this table:

- 1 duplicate row
- 1 missing value
- 1 invalid value
- 1 inconsistent label
- 1 outlier

Data Science Process

Step 2.2 — Data Preparation (cleaning)

Real-world data is messy — exploration is how these problems get found before modeling.

Issue	Found In Raw Data	Fix Applied	Result
Duplicate records	Borrower 04 duplicate row	Dropped the repeated row	14 rows → 13 rows
Inconsistent values	“New York” (Borrower 07)	Standardized to “NY”	1 canonical state code
Missing values	Borrower 11 income blank	Imputed with dataset median ($71,250)	No more blanks
Invalid data	Borrower 12 score = 900	Capped to the valid maximum, 850	Within 300–850 range
Outliers	Borrower 13 income = $1.25M	Flagged, not deleted — treated in Step 5	Kept for analysis

Data Science Process

Step 2.2 — Data Preparation (cleaning)

Real-world data is messy — exploration is how these problems get found before modeling.

Borrower ID	Credit Score	Annual Income	State	Interest Rate (%)
01	500	$42,000	NY	7.31
02	600	$58,000	CA	6.70
03	700	$71,000	TX	5.95
04	700	$71,500	TX	6.40
05	800	$95,000	FL	5.40
06	800	$102,000	FL	5.70
07	750	$88,000	NY	5.90
08	550	$39,000	CA	7.00
09	650	$61,000	TX	6.50
10	825	$110,000	FL	5.70
11	700	$71,250	NY	6.10
12	850	$67,000	CA	5.20
13	620	$1,250,000	TX	6.60

Data Science Process

Step 2.2 — Data Preparation (transforming)

Shaping data for the algorithm — types, scale, outliers, dimensionality, and sample size.

Data Types & Conversion

Numeric, categorical, or converted via encoding (e.g., poor=400) or binning.

Transformation

Normalization scales attributes (e.g., income vs. credit score) to a uniform 0–1 range.

Feature Selection

Reduce attributes without losing performance — mitigates the curse of dimensionality.

Data Sampling

A representative subset speeds up modeling; stratified sampling balances rare classes.

Training / Test Split

Rule of thumb: two-thirds training data, one-third held out for testing.

Data Science Process

Step 2.2 — Data Preparation (transforming)

Shaping data for the algorithm — types, scale, outliers, dimensionality, and sample size.

Data Types & Conversion

Numeric, categorical, or converted via encoding (e.g., poor=400) or binning.

Transformation

Normalization scales attributes (e.g., income vs. credit score) to a uniform 0–1 range.

State → Label Encoding

State	Code
NY	0
CA	1
TX	2
FL	3

ID	Credit Score	Normalized	Income (log)	Normalized
01	500	0.00	10.65	0.02
07	750	0.71	11.38	0.23
11	700	0.57	11.17	0.17
12	850	1.00	11.11	0.16
13	**620**	**0.34**	**14.04**	**1.00**

Data Science Process

Step 2.2 — Data Preparation (transforming)

Shaping data for the algorithm — types, scale, outliers, dimensionality, and sample size.

Feature Selection

Reduce attributes without losing performance — mitigates the curse of dimensionality.

Reduce attributes to what the model needs — without losing performance.

Kept

- credit_score_
- income_log
- interest_rate: the target variable

Dropped

- borrower_id — a unique identifier
- state text

Data Science Process

Step 2.2 — Data Preparation (transforming)

Shaping data for the algorithm — types, scale, outliers, dimensionality, and sample size.

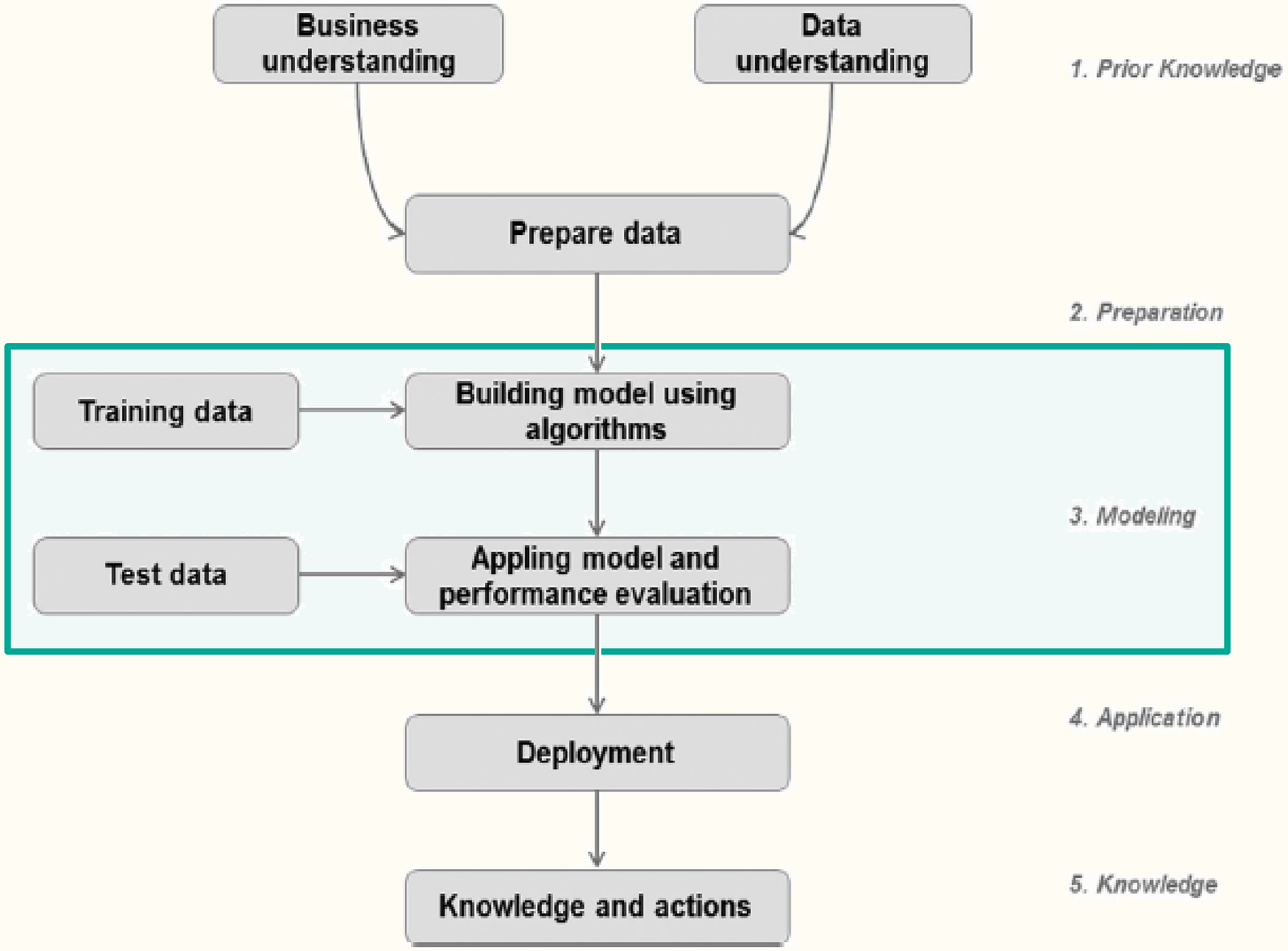
Borrower ID	Split
01	Train
02	Train
03	Train
04	Train
05	Train
06	Train
07	Train
08	Train
09	Train
10	Test
11	Test
12	Test
13	Test

13 clean records → 9 train · 4 test (69% / 31%)

Rule of thumb: roughly two-thirds training data, one-third held out for testing.



Data Science Process

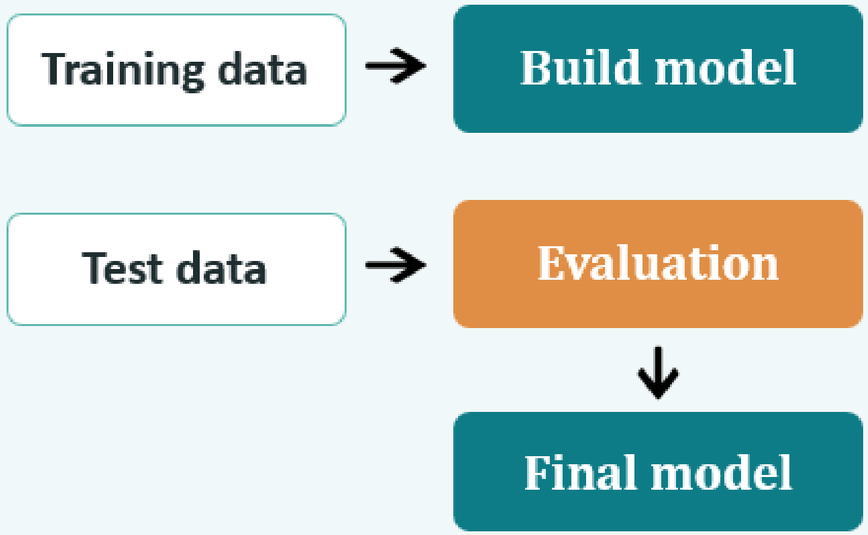


Data Science Process

2.3 Modeling

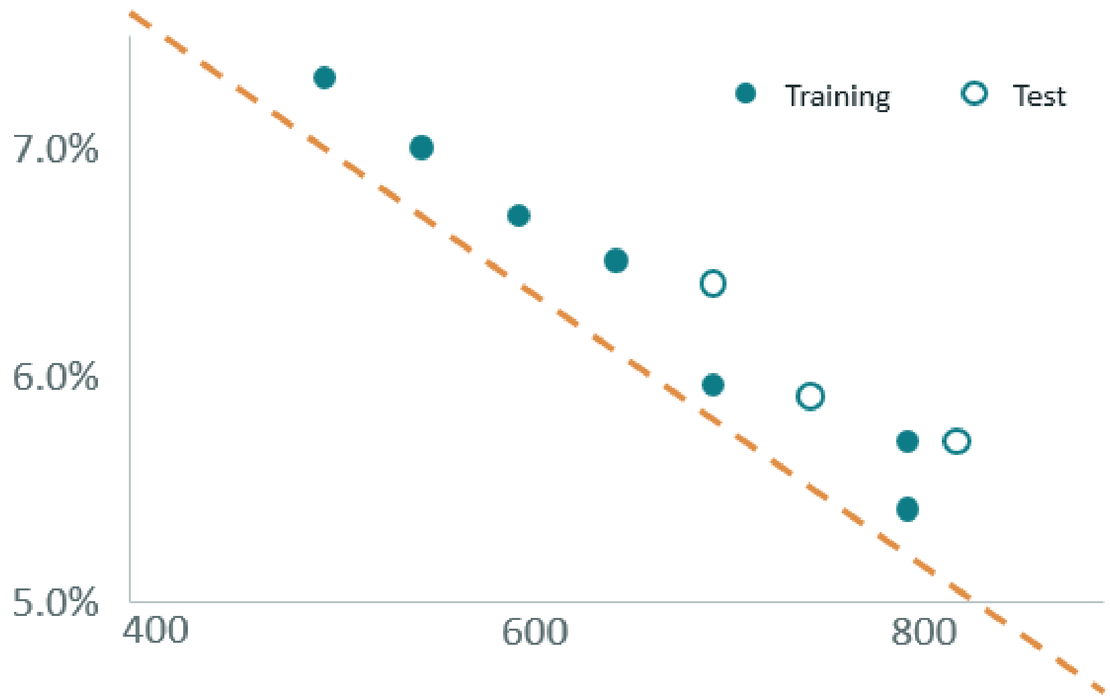
- *Building a representative model — abstracted from training data, validated against test data.*

Modeling steps



Classification and regression are predictive (need training + test data). Association analysis and clustering are descriptive — no target to test against.

Fitting a linear regression line

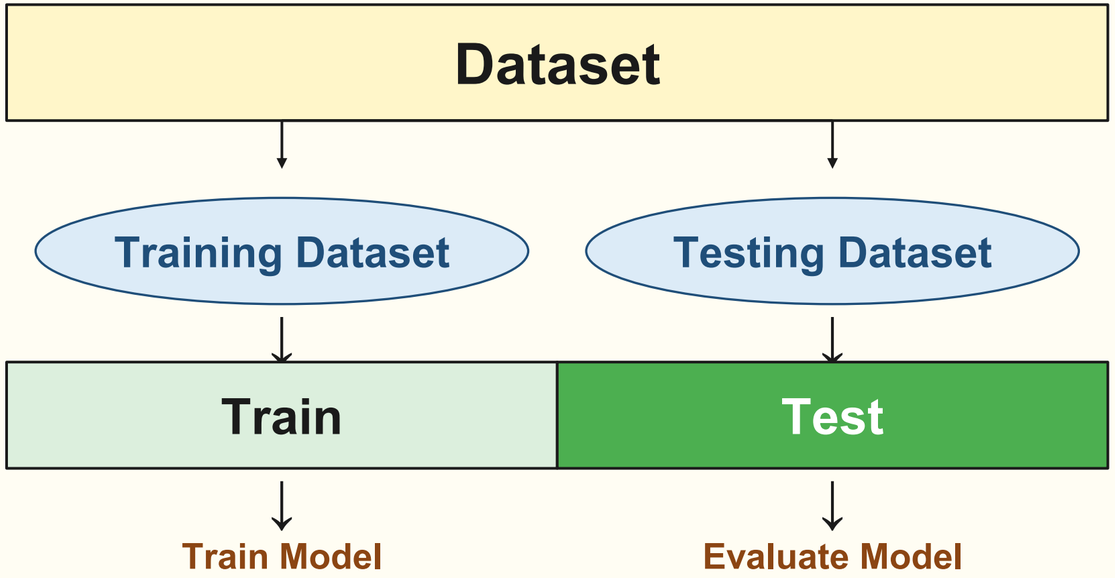


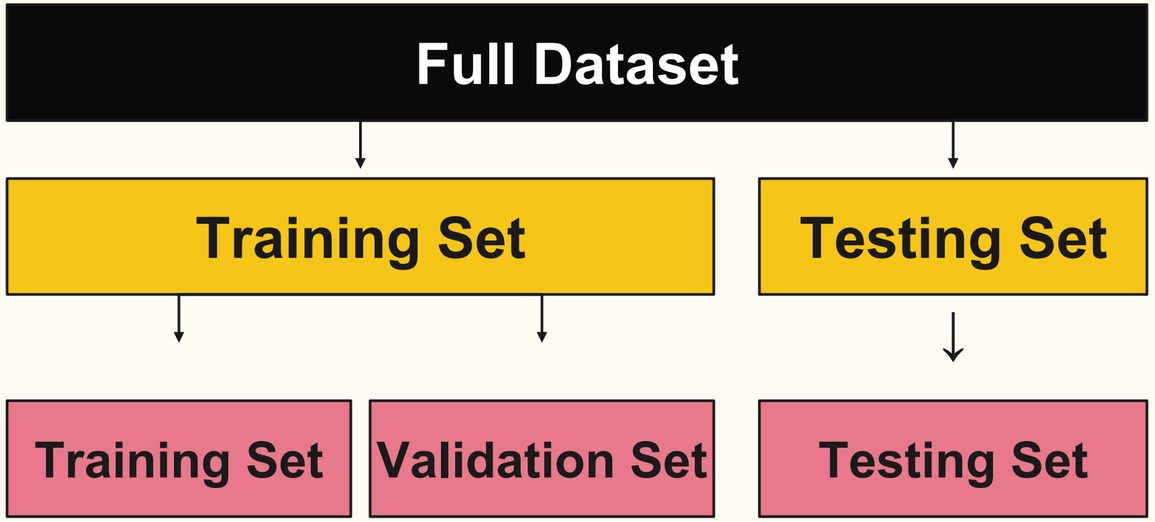
y = a·x + b → Interest rate = 10 – (6 × credit score) / 1000

Data Science Process

Evaluating Models

- *Comparing actual vs. predicted values on held-out test data — and combining models for better accuracy.*





Data Science Process

Why Cross-Validation Is Required

– *single train/test split can be misleading — cross-validation gives a more honest picture.*

The problem with one split



One split — one score, and it depends entirely on which records happen to land in the test set.

- A lucky split can make a weak model look great; an unlucky split can make a good model look bad
- Every record used for testing is a record the model never got to learn from
- Small datasets suffer most — losing 30% of records to testing is a real loss of signal
- There is no way to know how much the reported score would change with a different split

What cross-validation fixes



Every record takes a turn in the validation set — nothing is wasted.

- Uses the entire dataset for both training and validation, just not at the same time
- Averaging across k folds smooths out the effect of any one unusual split
- Produces a more stable, trustworthy estimate of how the model will perform on new data
- Makes overfitting easier to spot — a model that only works on one split will not score well across all of them

Data Science Process

Why Cross-Validation Is Required

- *single train/test split can be misleading — cross-validation gives a more honest picture.*

Cross-validation steps

k–1 folds (training) → **Build model**

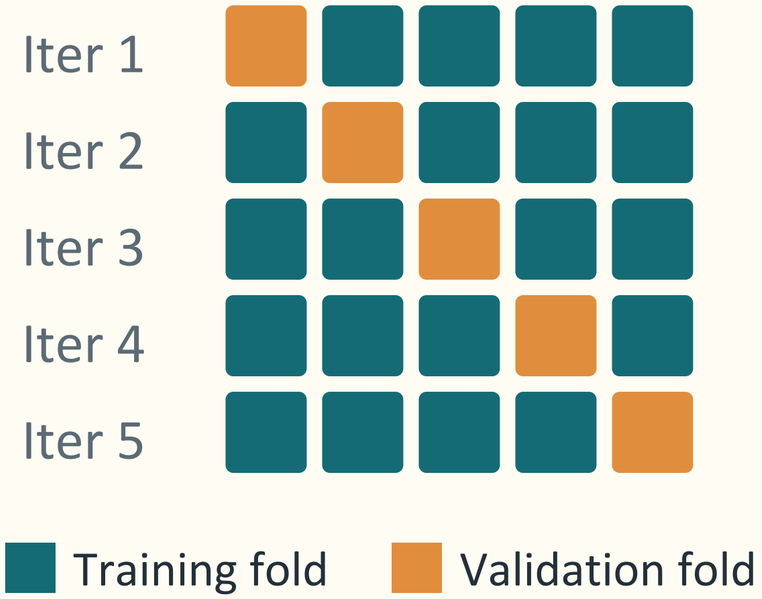
1-fold (validation) → **Evaluation**

↓ *repeat k times, rotating the held-out fold*

Average score

Cross-validation lowers the risk that one lucky (or unlucky) train/test split biases the evaluation — especially useful when the dataset is small.

5-fold rotation



$$\text{Accuracy}_{cv} = (1/k)\ \Sigma\ \text{Accuracy}_{i}$$

Data Science Process

Applying Cross-Validation to the Original Dataset

- *Using the 10-record credit score / interest rate dataset introduced earlier, with k = 5.*

Step 1 — assign each record to a fold

Borrower	Credit Score	Interest Rate	Fold
01	500	7.31%	Fold 1
02	600	6.70%	Fold 1
03	700	5.95%	Fold 2
04	700	6.40%	Fold 2
05	800	5.40%	Fold 3
06	800	5.70%	Fold 3
07	750	5.90%	Fold 4
08	550	7.00%	Fold 4
09	650	6.50%	Fold 5
10	825	5.70%	Fold 5

Step 2 — rotate the validation fold

Validate: Fold 1	Borrowers 01, 02 held out; the other 8 borrowers train the model.
Validate: Fold 2	Borrowers 03, 04 held out; the other 8 borrowers train the model.
Validate: Fold 3	Borrowers 05, 06 held out; the other 8 borrowers train the model.
Validate: Fold 4	Borrowers 07, 08 held out; the other 8 borrowers train the model.
Validate: Fold 5	Borrowers 09, 10 held out; the other 8 borrowers train the model.

After 5 iterations, average the 5 validation errors for one overall cross-validated performance estimate.

Data Science Process

Cross Validation Techniques

- *Different ways to split data, depending on dataset size, balance, and structure.*

Hold-Out Validation

A single split into train and test sets (e.g. 70/30). Fast, but the result depends heavily on which split you happen to get.

k-Fold Cross-Validation

Data is split into k folds; each fold takes a turn as the validation set. The most common general-purpose method.

Stratified k-Fold

Like k-fold, but each fold preserves the original class proportions. Best for imbalanced classification problems.

Leave-One-Out (LOOCV)

Each single record is held out as validation, once each. Uses almost all data for training, but is costly on large datasets.

Leave-P-Out

A generalization of LOOCV where p records are held out at a time instead of just one.

Data Science Process

Cross Validation Techniques — Hold-Out Validation

- *One split, one score — the simplest and fastest way to check a model.*

How it works

- Shuffle the dataset, then split it once into a training set and a test set
- Common ratios: 70/30 or 80/20, train/test
- Train the model on the training portion only
- Score it once on the held-out test portion — that score is the final estimate

Trade-off

Fast and simple, but the score depends heavily on which records ended up in the test set — no averaging to smooth that out.

Worked example — 10 borrowers, 70/30 split



Train: borrowers 01–07 (70%) **Test:** borrowers 08–10 (30%)

The model is built once on borrowers 01–07 and scored once on 08–10. If borrower 10 (an unusual high-credit-score case) happens to land in test, the score may look different than if it had been in training.

Data Science Process

Cross Validation Techniques — k-Fold Cross-Validation

- *Every record trains the model in some iterations and validates it in exactly one.*

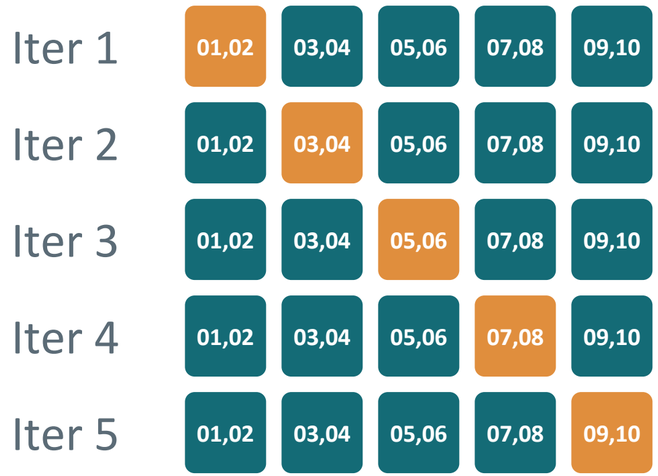
How it works

- Split the dataset into k equal-sized folds (k = 5 is common)
- For each of the k iterations, hold out one-fold for validation
- Train on the remaining k–1 folds, then score on the held-out fold
- Average the k scores for the final cross-validated estimate

Trade-off

More reliable than a single hold-out split, at the cost of training the model k times instead of once.

Worked example — 10 borrowers, k = 5



Each cell shows which two borrowers sit in that fold. Iteration 3, for example, validates on borrowers 05 & 06 and trains on the other 8.

Data Science Process

Cross Validation Techniques — Stratified k-Fold

– *Keeps the same class balance in every fold — critical when classes are uneven.*

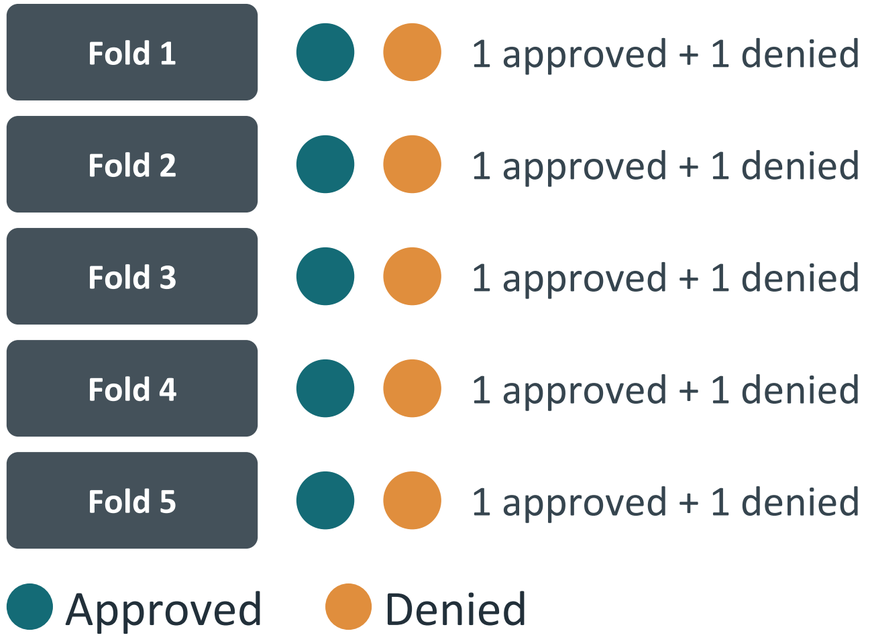
How it works

- Look at the proportion of each class label in the full dataset
- Build each of the k folds so it mirrors that same proportion
- Run standard k-fold cross-validation on top of these balanced folds
- Prevents a fold from accidentally containing only one class

Trade-off

Same cost as k-fold, but essential for imbalanced problems like fraud or churn, where the minority class can otherwise vanish from a fold entirely.

Worked example — 10 loan applications (5 approved, 5 denied)



Every fold keeps the original 50/50 approved-to-denied ratio, so each validation pass sees a realistic mix of both outcomes.

Data Science Process

Cross Validation Techniques — Leave-One-Out (LOOCV)

- *The most exhaustive form of k-fold: k equals the number of records.*

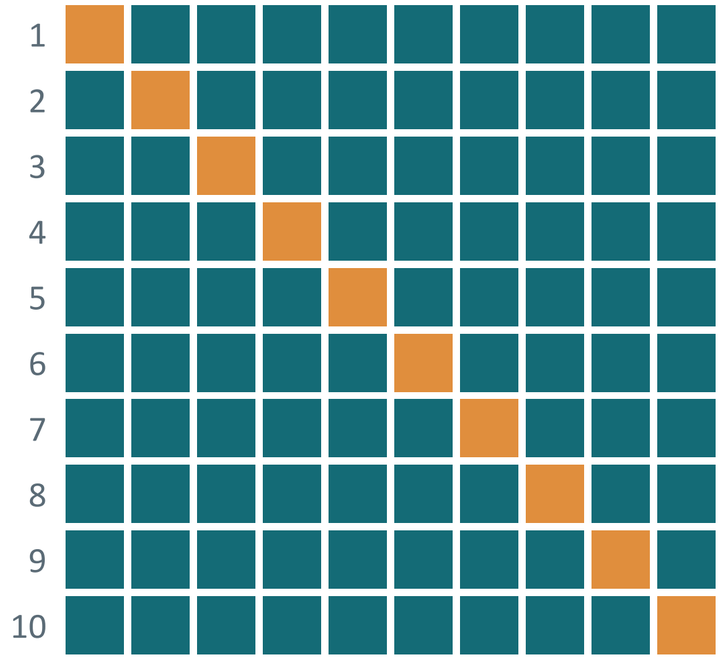
How it works

- For a dataset of n records, run n iterations
- Each iteration holds out exactly one record for validation
- Train on the remaining n–1 records every time
- Average all n validation scores for the final estimate

Trade-off

Uses almost all the data for training every time, which is great for very small datasets — but n models is expensive once the dataset grows large.

Worked example — 10 borrowers → 10 iterations



Row = iteration, column = borrower (01–10). The single orange cell in each row is the one borrower validated that round; all others train the model.

Data Science Process

Cross Validation Techniques — Leave-P-Out

- *A generalization of LOOCV — hold out every possible group of p records.*

How it works

- Choose p, the number of records to hold out per iteration
- Form every possible combination of p records from the dataset
- Each combination becomes one validation set; the rest is training
- Average the scores across all C(n, p) combinations

Trade-off

The number of combinations grows fast — C(n, p) can become huge, so Leave-P-Out is only practical on small datasets.

Worked example — 10 borrowers, p = 2

C(10, 2) = 45 possible validation pairs

Iteration 1: hold out {01, 02} — train on the other 8

Iteration 2: hold out {01, 03} — train on the other 8

Iteration 3: hold out {01, 04} — train on the other 8

… continues through every possible pair …

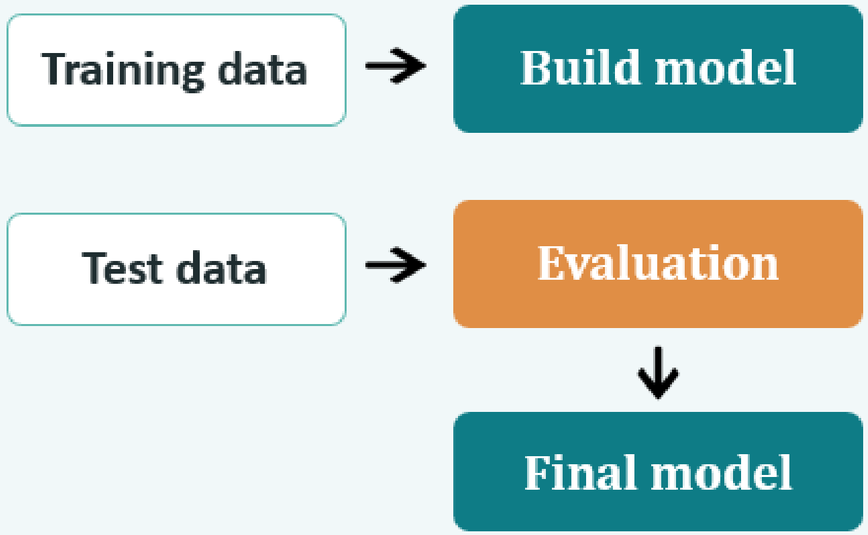
Iteration 45: hold out {09, 10} — train on the other 8

Data Science Process

2.3 Modeling

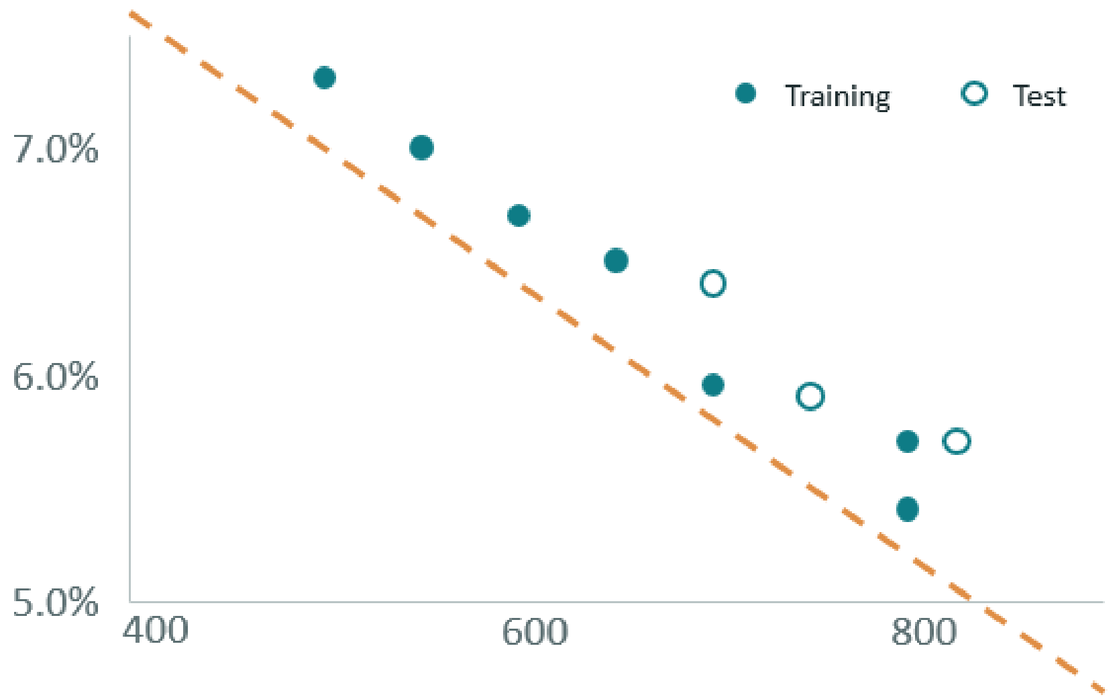
- *Building a representative model — abstracted from training data, validated against test data.*

Modeling steps



Classification and regression are predictive (need training + test data). Association analysis and clustering are descriptive — no target to test against.

Fitting a linear regression line



y = a·x + b → Interest rate = 10 – (6 × credit score) / 1000

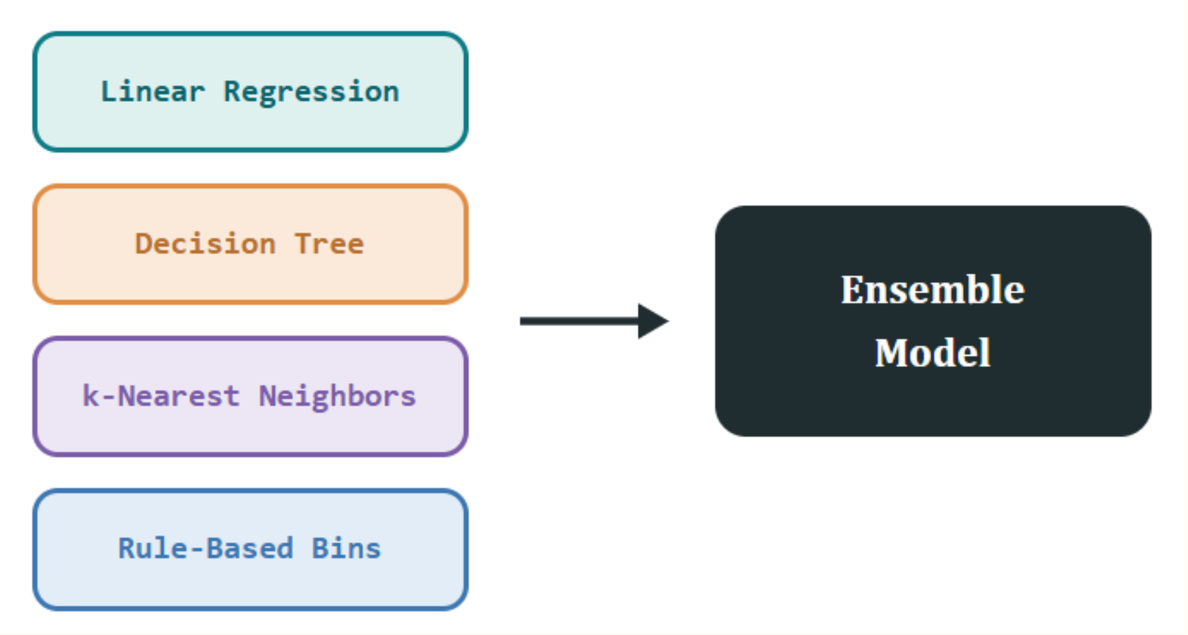
Data Science Process

Combining Models

- *Comparing actual vs. predicted values on held-out test data — and combining models for better accuracy.*

Ensemble Modeling

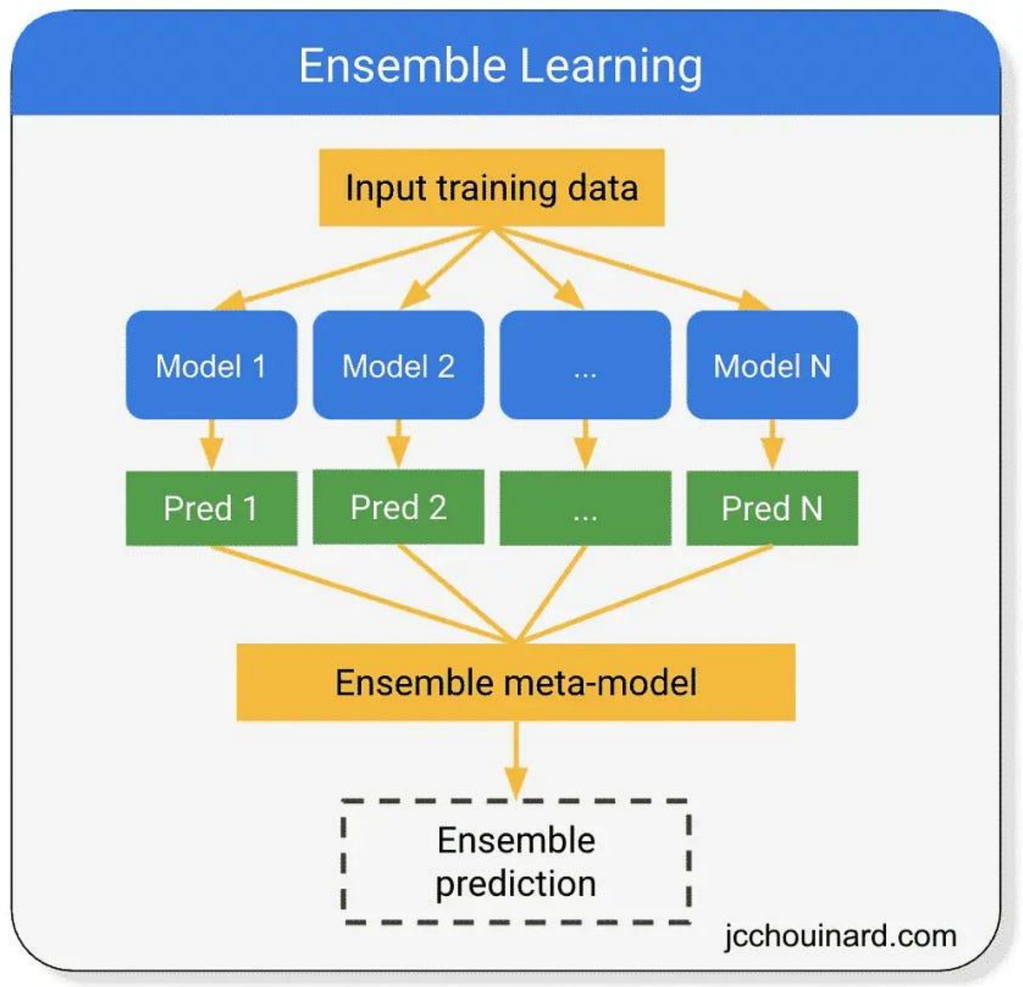
- Multiple diverse, independent base models are combined to predict one outcome.
- The combined “meta model” is called the ensemble model.
- As long as base models are diverse, prediction error decreases — the “wisdom of crowds.”
- Though built from many models, an ensemble acts and performs as a single model. Most practical data science applications use this technique.



Data Science Process

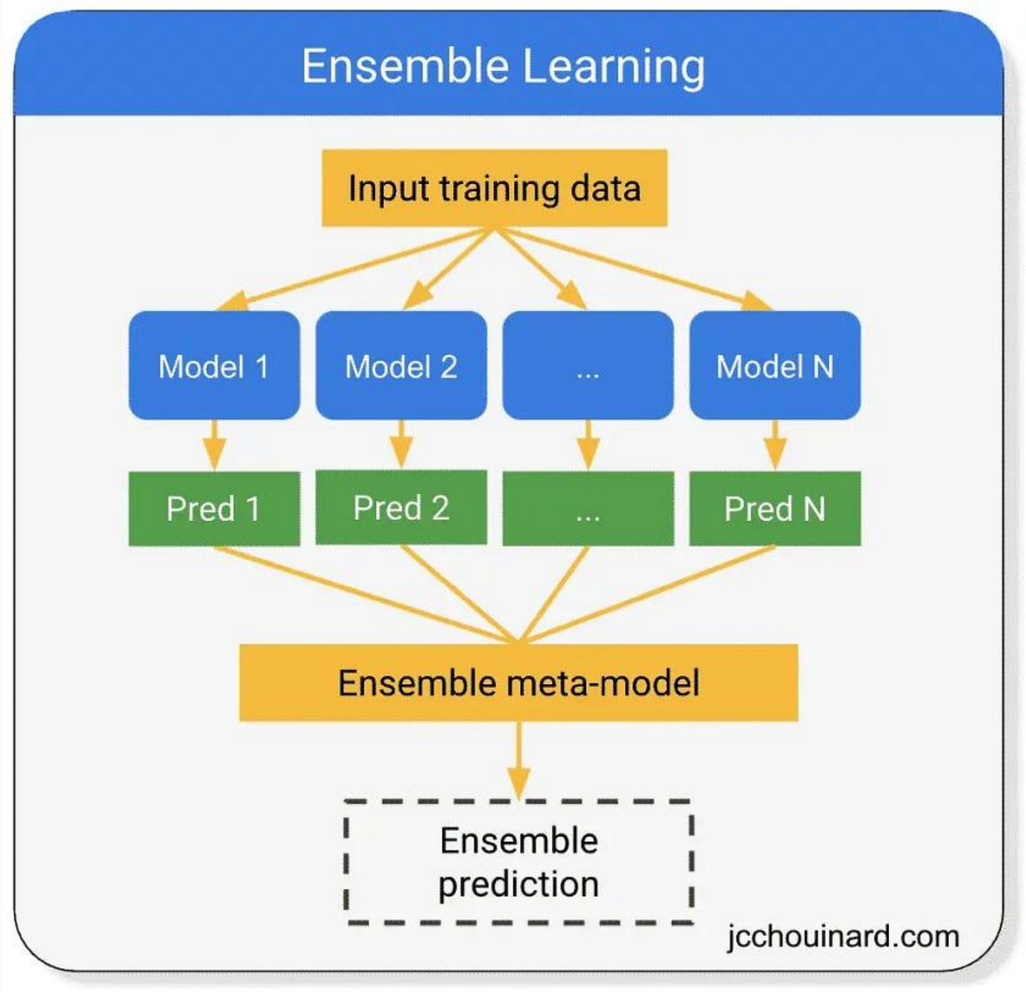
Ensemble Learning

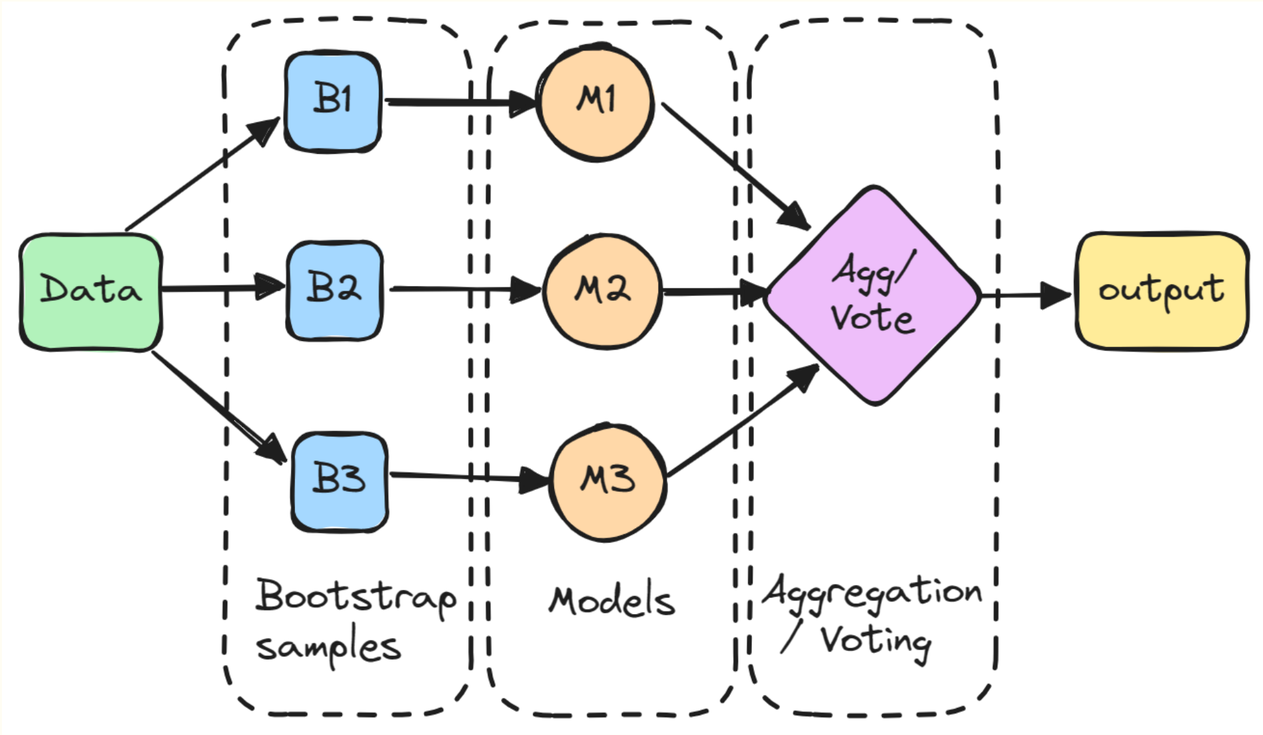
- Ensemble Learning is a technique that combines multiple machine learning models to improve overall predictive performance. The basic idea is that a group of weak learners can come together to form one strong learner.
- An ensemble model typically consists of two steps:
- Multiple machine learning models are trained independently.
- Their predictions are aggregated in some way, such as by voting, averaging, or weighting. This ensemble is then used to make the overall prediction.



Data Science Process

Ensemble Learning

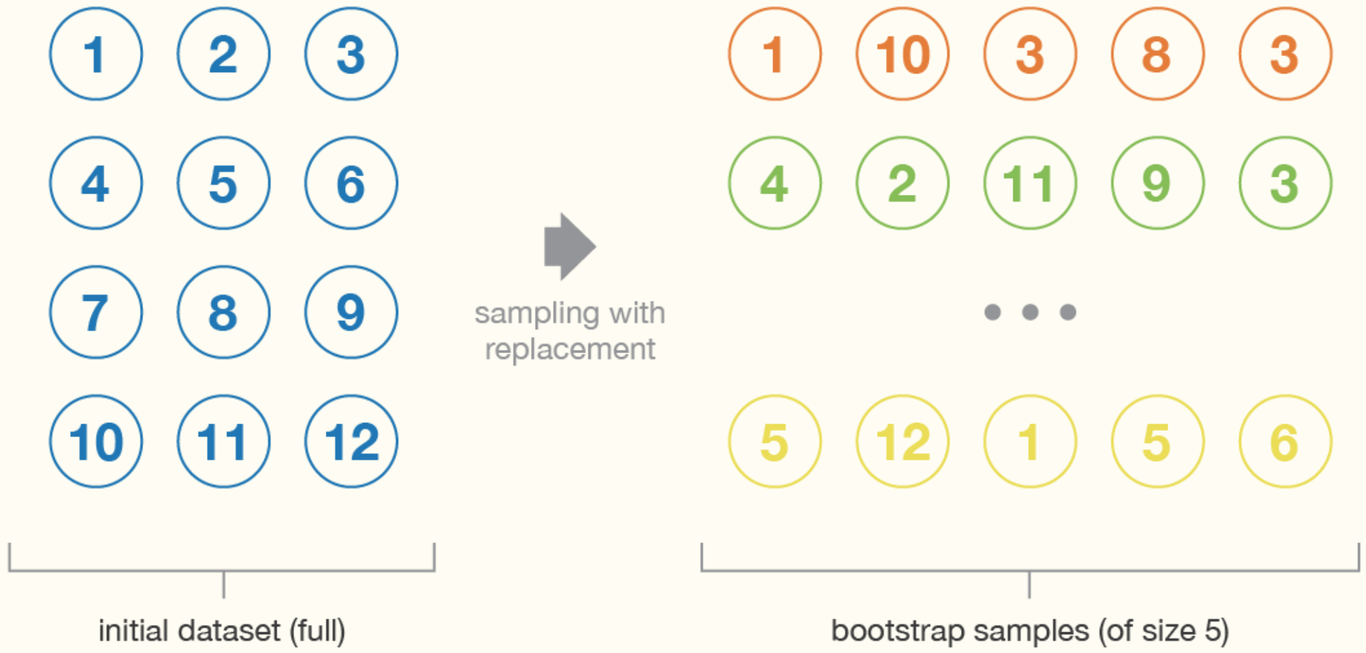




Data Science Process

Bootstrapping:

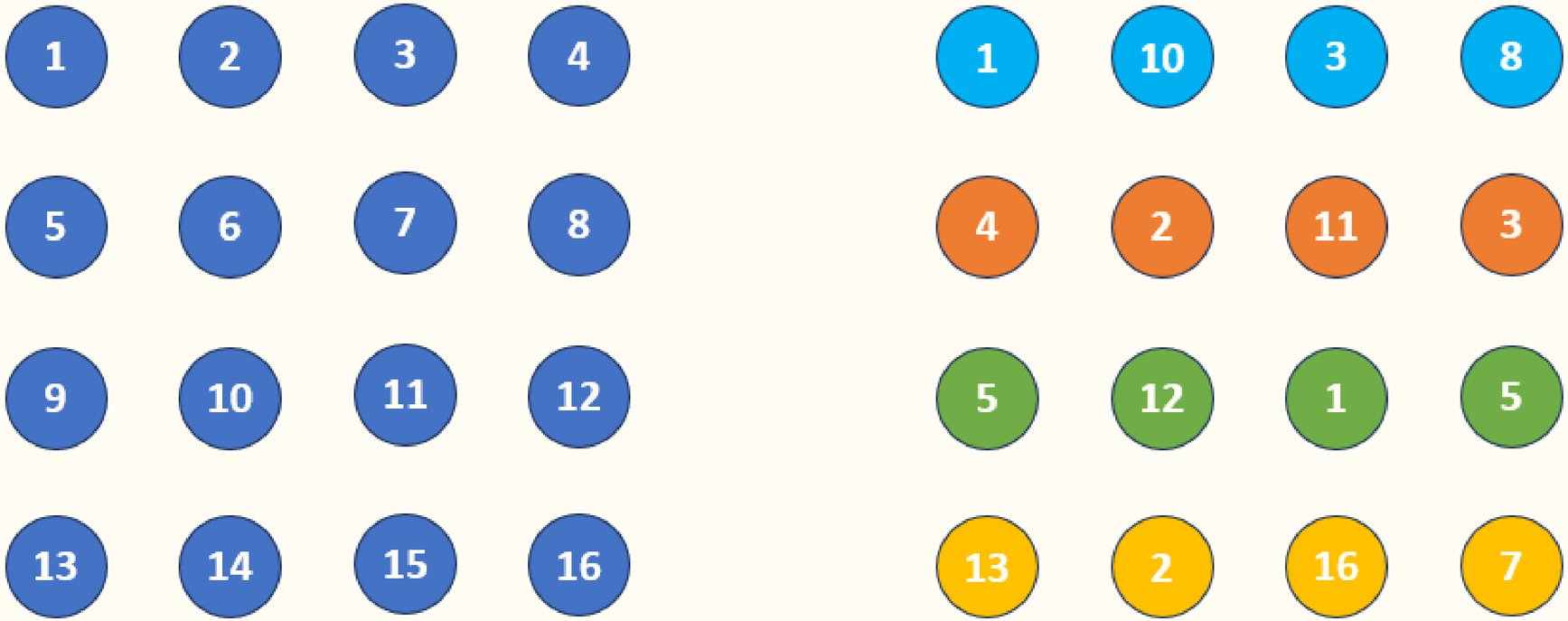
- Bootstrapping is the process of generating bootstrapped samples from the given dataset. The samples are formulated by randomly drawing the data points with replacement



Data Science Process

Bootstrapping:

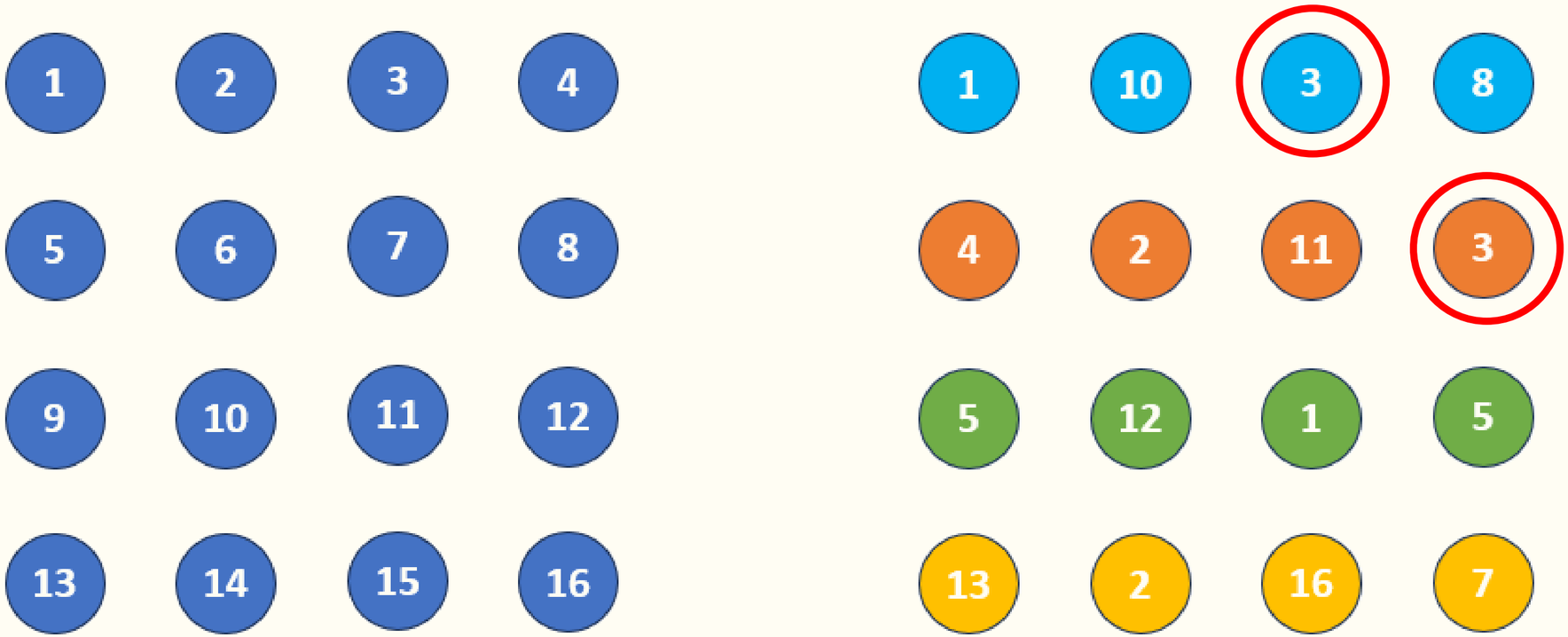
- Bootstrapping is the process of generating bootstrapped samples from the given dataset. The samples are formulated by randomly drawing the data points with replacement
- Bootstrap samples with replacement. The size of the sample is usually the same as the original dataset.



Data Science Process

Bootstrapping:

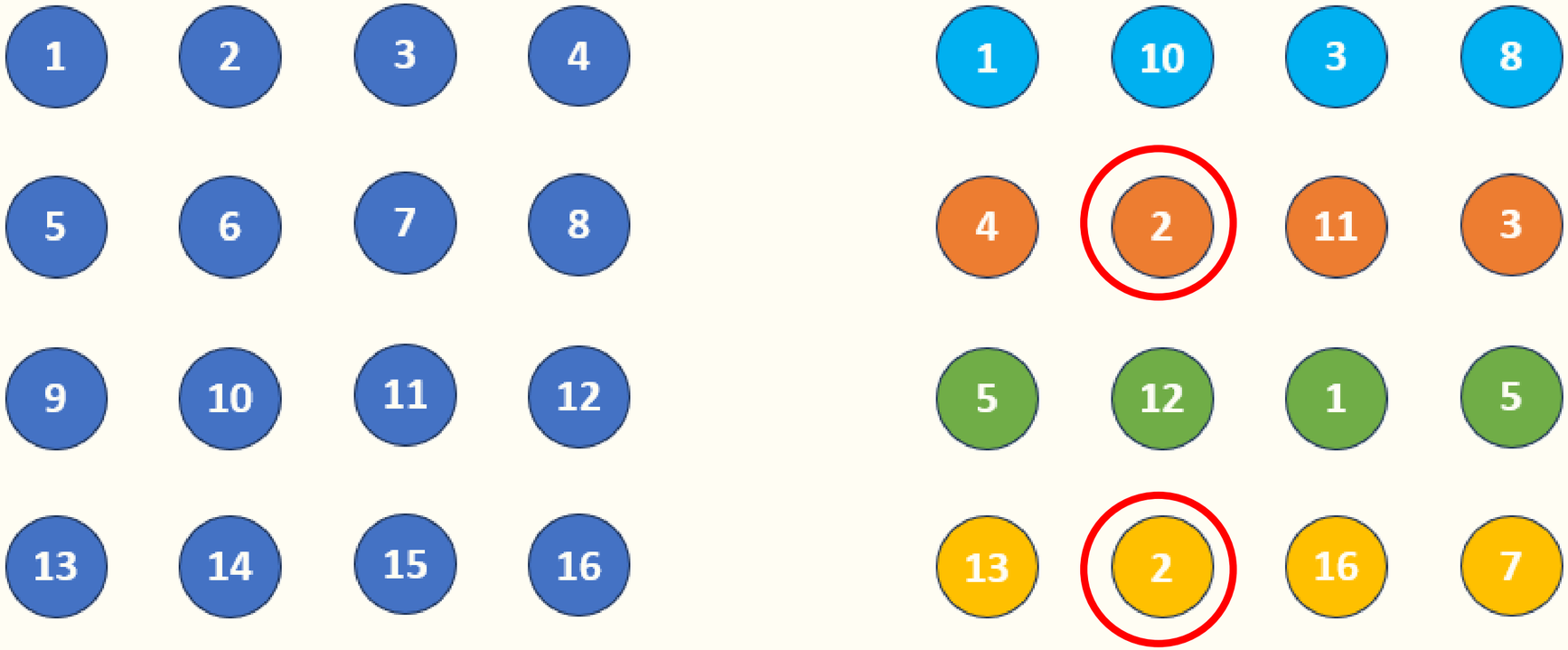
- Bootstrapping is the process of generating bootstrapped samples from the given dataset. The samples are formulated by randomly drawing the data points with replacement
- Bootstrap samples with replacement. The size of the sample is usually the same as the original dataset.



Data Science Process

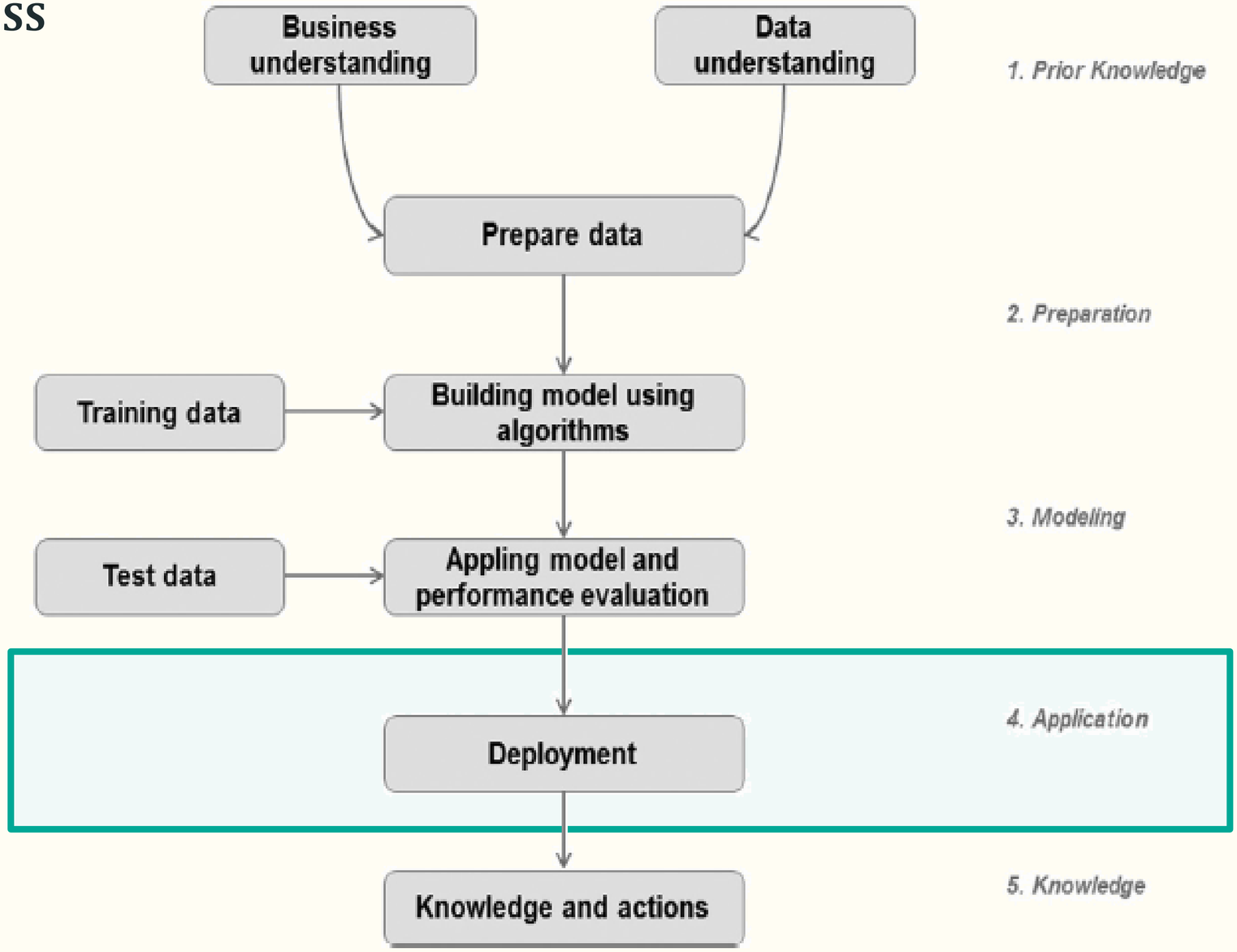
Bootstrapping:

- Bootstrapping is the process of generating bootstrapped samples from the given dataset. The samples are formulated by randomly drawing the data points with replacement
- Bootstrap samples with replacement. The size of the sample is usually the same as the original dataset.



Data Science Process

THE DATA SCIENCE PROCESS



Data Science Process

2.4 Application

- *Deployment is where a model becomes production-ready — assessed across five dimensions.*

Production Readiness

Real-time credit decisions vs. batch-processed customer segmentation — the use case sets the bar.

Technical Integration

Models port via PMML or by invoking the data science tool directly inside production applications.

Response Time

k-NN is slow to predict but fast to build; decision trees are the reverse — trade-offs must be weighed.

Model Refresh

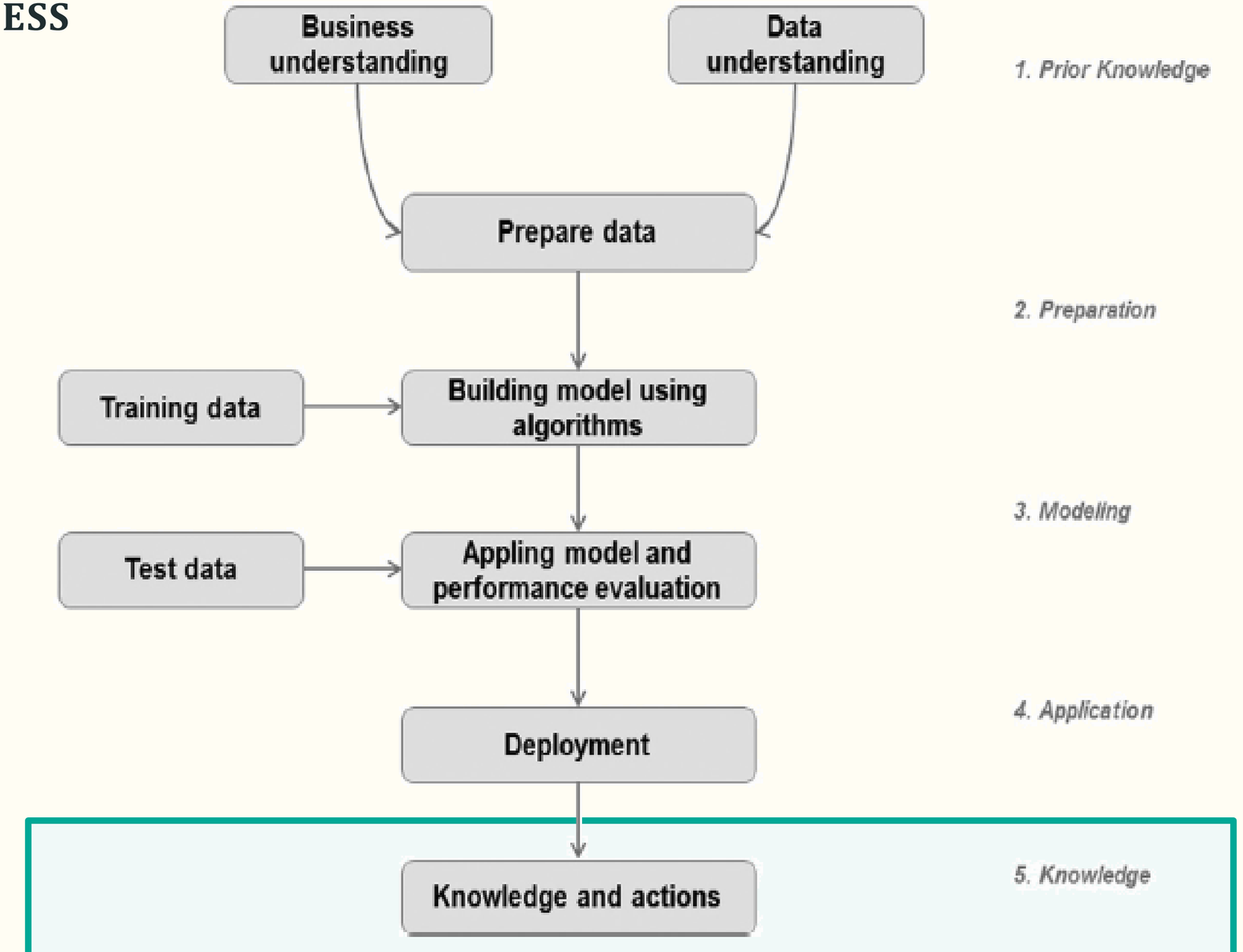
Conditions change after deployment; test error routinely and refresh the model when it exceeds a threshold.

Assimilation

Translate findings into organizational knowledge — quantify risk and business impact, not just the model.

Data Science Process

THE DATA SCIENCE PROCESS



Data Science Process

2.5 Knowledge

- *From prior knowledge to posterior knowledge — the incremental insight gained.*

- Not all discovered patterns lead to incremental knowledge — it's up to the practitioner to invalidate spurious, irrelevant signals.
- The process is a framework to invoke the right questions and guide, through the right approaches, a solution.
- It is not a set of rigid rules, but a set of iterative, distinct steps that aid in knowledge discovery.

Data Science Process

Small Dataset Walkthrough (1/2): Understand → Prepare

Telecom customer churn prediction — trying to predict whether a customer will cancel their subscription

1 · Business & Data Understanding

Raw Data

Customer	Usage (GB/mo)	Tenure (mo)	Support Calls	Churned ?
C001	12	24	1	No
C002	3	5	4	Yes
C003	18	36	0	No
C004	2	3	5	Yes
C005	9	15	2	?

- *Objective- predict whether a customer will cancel their subscription*
- *Understanding Telecom company's business.*
- *Understanding raw data, attribute relation*

2 · Preparation

Prepared Data

Customer	Usage (z-score)	Tenure (mo)	Support Calls	Churn
C001	+0.35	24	1	0
C002	-0.98	5	4	1
C003	+1.25	36	0	0
C004	-1.10	3	5	1
C005	-0.05	15	2	?

- *Cleaning applied: usage scaled to a z-score, churn recoded as 0/1, no missing values remain.*
- *C005 hasn't been labeled yet — that's the row we'll hold out to test the model.*

Data Science Process

Small Dataset Walkthrough (2/2): Model → Act

3 · Modeling — Training Data

Customer	Support Calls	Churn
C001	1	0
C002	4	1
C003	0	0
C004	5	1

Learned rule:
Support Calls ≥ 4 → predict Churn

3 · Modeling — Testing Data

Customer	Support Calls	Predicted	Actual
C005	2	**No churn**	**No churn**

✓Prediction matches — 1 correct out of 1 test row

4 · Application — Deployment

The rule is wired into the CRM: any customer whose support-call count crosses 4 is flagged automatically, in real time.

5 · Knowledge & Actions

Flagged customers get a proactive retention call and a loyalty discount before they consider leaving.

Syllabus Contents

1.5 | Difference between data science and data analytics

LO 1.1 - Apply the knowledge of discrete structures, linear algebra, statistics, and numerical techniques to interpret fundamental data characteristics such as volume, dimensionality, and complexity. (PI 1.1.1)

LO 1.3- Evaluate a given data-related problem statements and determine the required analysis, while continually learning and adapting to new data science tools and techniques

Reference: Vijay Kotu, Bala Deshpande. Data Science Concepts and Practice||, Elsevier, M.K. Publishers.

Data science and Data analytics

Aspect	Data Science	Data Analytics
Meaning	A broad field that uses scientific methods, algorithms, and systems to extract knowledge from data	Focuses on examining data to find patterns, trends, and insights
Scope	Very broad (includes analytics, ML, AI, statistics, big data)	Narrower (subset of data science)
Main Goal	Predict future outcomes and build intelligent systems	Understand past and present data to support decisions
Type of Analysis	Descriptive, Diagnostic, Predictive, Prescriptive	Mostly Descriptive and Diagnostic
Techniques Used	Machine Learning, Deep Learning, Statistics, Data Mining	Statistical analysis, querying, visualization
Data Types	Structured, Semi-structured, Unstructured (text, images, audio)	Mostly Structured data
Tools	Python, R, TensorFlow, PyTorch, Spark	Excel, SQL, Power BI, Tableau
Programming	Strong programming skills required	Basic to moderate programming
Outcome	Models, predictions, automated decision systems	Reports, dashboards, insights
Example Question	Will customers stop using our service next month?	Why did sales drop last quarter?

Data science and Data analytics

Data Science vs. Data Analytics

Two related but distinct disciplines for working with data

DA Data Analytics

Examines existing data to answer specific questions and explain what has already happened.

FOCUS

Descriptive & diagnostic — “what happened & why”

TOOLS

Excel, SQL, Tableau, Power BI

Example: A retailer looks at last month’s sales report and finds that umbrella sales spiked every time it rained.

DS Data Science

Builds models on data (often including analytics as a step) to predict future outcomes and automate decisions.

FOCUS

Predictive & prescriptive — “what will happen & what to do”

TOOLS

Python, R, Machine Learning, AI

Example: The same retailer builds a model using weather forecasts to predict next week’s umbrella demand and auto-adjust stock.

Data science and Data analytics

Simple, Everyday Examples

Netflix

ANALYTICS Netflix looks at what you've already watched and shows it in "Continue Watching."

SCIENCE Netflix predicts what you'll enjoy next and recommends it before you even search.

School Grades

ANALYTICS A teacher checks last month's test scores to see who struggled with maths.

SCIENCE A teacher uses a model to predict who might fail next term and helps them early.

Weather & Umbrella

ANALYTICS You look at yesterday's weather and see that it rained.

SCIENCE A weather app predicts it will rain tomorrow, so you carry an umbrella today.

Cricket

ANALYTICS A coach looks at the last match's scorecard to see who scored the most runs.

SCIENCE A coach uses player data to predict who will perform best tomorrow and picks the team.

Data science and Data analytics

DATA ANALYTICS

HEALTHCARE

A hospital reviews last quarter’s records and finds readmission rates were highest among patients discharged on Fridays.

MARKETING

A marketing team checks campaign reports and sees email open rates were 20% higher on Tuesday mornings.

FINANCE

An accountant reviews monthly expense sheets and identifies which department overspent its budget.

DATA SCIENCE

HEALTHCARE

A hospital builds a model that flags high-risk patients on admission so it can schedule follow-up care in advance.

MARKETING

A marketing team trains a model to predict which customers are likely to churn and auto-sends them a retention offer.

FINANCE

A bank builds a machine learning model to detect and block fraudulent transactions in real time.

Syllabus Contents

1.6 Data Science Applications

LO 1.3- Evaluate a given data-related problem statements and determine the required analysis, while continually learning and adapting to new data science tools and techniques. (PI 2.1.1, PI 11.1.1)

Reference: Vijay Kotu, Bala Deshpande. Data Science Concepts and Practice||, Elsevier, M.K. Publishers.

Data science Applications

Healthcare

Predict, diagnose, and personalize care

Disease prediction

Models flag diabetes or heart-disease risk from patient history and vitals

Medical imaging

Deep learning detects tumors in X-rays, MRIs and CT scans

Drug discovery

ML screens millions of compounds to speed up new treatments

Example: Hospital AI systems analyze imaging and patient records to detect early-stage cancers, often matching or exceeding radiologist accuracy in specific tasks.

Data science Applications

Finance & Banking

Detecting fraud, managing risk, and automating trades

Fraud Detection

Anomaly-detection models flag suspicious transactions in real time

Algorithmic Trading

Models analyze market data to execute trades in milliseconds

Credit Scoring

Predictive models assess loan default risk from applicant data

Data science Applications

Retail & E-Commerce

Recommendation Engines

Collaborative filtering & content-based models suggest products based on browsing and purchase history, driving a large share of online sales.

Example: Amazon's "customers who bought this also bought" feature is powered by collaborative-filtering algorithms.

Demand Forecasting

Predict inventory needs by store and season to reduce stockouts

Dynamic Pricing

Algorithms adjust prices in real time based on demand and competitors

Customer Segmentation

Clustering groups shoppers for targeted marketing campaigns

Data science Applications

Transportation & Logistics

Optimizing routes, predicting demand, and enabling autonomy

Autonomous Vehicles	Computer vision and sensor-fusion models let self-driving cars perceive roads and make real-time decisions.
Route Optimization	Algorithms compute the fastest, cheapest delivery routes across thousands of stops daily.
Demand Prediction	Ride-hailing apps forecast rider demand to balance driver supply and dynamic pricing.

Data science Applications

Marketing & Social Media

Targeted Advertising

Behavioral models match ads to users most likely to engage or convert

Sentiment Analysis

NLP models gauge public opinion about brands from posts and reviews

Churn Prediction

Models identify customers likely to unsubscribe, enabling retention offers

Chatbots

Conversational AI handles customer queries and support at scale

Example: Netflix and Spotify-style engines aside, brands like Sephora use purchase and browsing data to personalize email and app promotions per shopper.

Data science Applications

Entertainment & Media

Personalization at massive scale

- **Content Recommendation**

  Models rank and surface shows, movies, or songs based on individual viewing/listening habits

- **Automated Tagging**

  Computer vision tags scenes, actors and content for better search and moderation

- **Churn & Retention**

  Predictive models flag subscribers at risk of canceling to trigger win-back offers

Netflix

Its recommendation algorithm analyzes viewing patterns across 280M+ subscribers, influencing over three-quarters of what people watch.

Syllabus Contents

1.1 Introduction to Data Science

1.2 Motivation to use Data Science Techniques: Volume, Dimensions and Complexity

1.3 Data Science Tasks and Examples

1.4 Data Science Process, Model building, Cross Validation, Bootstrapping

1.5 Difference between data science and data analytics

1.6 Applications of Data Science

Activity Time

Pick a challenge and put your data science thinking to the test

01

Is This a Data Science Problem?

Sort real-world scenarios into data science vs. not — sharpen your instinct for what the field actually covers.

02

Find the Spurious Correlation

Spot the correlations that look meaningful but fall apart under scrutiny.

03

Data Science Escape Room

Work through a linked set of data puzzles to unlock the next clue and escape the room.

04

Analytics Looks Back · Science Looks Ahead

Crack each case: decide whether it's explaining the past or predicting what's next.